Componentwise saturation-height bounds for relative class-number growth in the cyclotomic $\mathbb{Z}_2$- and $\mathbb{Z}_3$-towers

Blueprint: statements, dependency graph and full proofs

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Trust legend. Every node title ends with its label: [F] Lean 4 kernel, toolchain v4.31.0-rc1, Mathlib d568c8c0, `#print axioms` within [propext, Classical.choice, Quot.sound]; [C] certificate accepted by a read-only checker that replays it from the definitions: `scripts/family_verify.sage` (sha256 0f38bb0d) for the $p = 2$ family, `scripts/verify_p3_readonly.sage` for the $p = 3$ covolumes (verifier step 04f); [L] published theorem used as an explicit hypothesis, verbatim location in the SOURCES block of `content.tex`; [M] proved in this document and in the paper, not machine-checked; [E] experiment, used in no theorem. Composite labels name the core and the inputs. Every F and M node carries a complete human-readable proof (single-source files `proofs/*.tex`); `\leanok` marks kernel-checked statements only. Package: weber_general_n 1.0.1. This repository copy carries the round only; the manifest checksum and the local verifier run are recorded in `RELEASE_STATUS.md`, while the commit SHA, the CI run and the tag live outside the source tree, in the release metadata and the CI attestation asset (a document cannot contain its own hash; GPT r14 sect 7). The two experiment nodes (`exp:class1`, `exp:svp`) are deliberately isolated: no theorem uses them.

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Chapter 1

Setting

Definition 1.1 (Weber layers, relative units, the KY unit, components)**.** $B_n = \mathbb{Q}(X_n)$, $X_n = 2\cos(2\pi/2^{n+2})$; $G_n = \langle\sigma\rangle$, $\sigma(X_n) = 2\cos(3\cdot 2\pi/2^{n+2})$; $\tau = \sigma^{2^{n-1}}$; E_n the units, $RE_n^+ = \{\varepsilon : N_{n/n-1}\varepsilon = 1\}$; $\varepsilon_n = (X_n+1)/(X_n-1)$; $A_n = \langle \pm 1, \varepsilon_n\rangle_{\mathbb{Z}[G_n]}$; $m = 2^{n-1}$; $R_n = \mathbb{Z}[x]/(x^m+1)$; $u_a = \prod \sigma^i(\varepsilon_n)^{a_i}$; $\lambda_j = \log|\sigma^j\varepsilon_n|$; $H_n : \mathbb{Z}^m \to \mathbb{R}^{2m}$, $(H_n a)_j = \sum_i a_i\lambda_{i+j}$; $D_n = \operatorname{covol} H_n(\mathbb{Z}^m)$; $L_n = \sqrt{2^n}\log(2+\sqrt{5})$; $d_n(\ell) = \operatorname{ord}_{2^n}(\ell)$; for $f \mid x^m+1$ irreducible over $\mathbb{F}_\ell$: $M_f = ((x^m+1)/f)\mathbb{F}_\ell[x]/(x^m+1)$, $L_f = \pi^{-1}(M_f)$, $\Lambda_f = \ell^{-1}H_n(L_f)$; $C_n = (2/\pi)^{m/2}\Gamma(2+m/2)D_n/L_n^m$. The KY identification (KY sect. 4) is

$$R_n/\ell R_n = \mathbb{F}_\ell[x]/(x^m+1) \xrightarrow{\sim} A_n^{1/\ell}/A_n, \qquad \bar{a} \mapsto r_a A_n, \quad r_a := \prod_{i<m} \sigma^i(\varepsilon_n)^{a_i/\ell}, \tag{1.1}$$

$A_n^{1/\ell} = \{x \in \mathbb{R} : x^\ell \in A_n\}$, r_a the real ℓ-th root of u_a; well defined on classes mod ℓ since $a \mapsto a + \ell b$ multiplies r_a by $u_b \in A_n$.

Chapter 2

Literature inputs

Lemma 2.1 (KY index [L]). $[RE_n^+ : A_n] = h_n/h_{n-1} =: k_n$. *(KY Eq. (17), p.10, exact equality.)*

Lemma 2.2 (KY freeness and identification [L]). *$A_n/\{\pm 1\}$ is free abelian on $\sigma^i(\varepsilon_n)$, $0 \le i < m$; $R_n/\ell R_n \simeq A_n^{1/\ell}/A_n$, $\bar{a} \mapsto r_a A_n$ with r_a the real ℓ-th root of u_a. (KY sect. 4, pp.13–14.)*

Lemma 2.3 (KY saturation [L]). *If $\ell \mid k_n$ then some irreducible $f_0 \mid x^m + 1$ over $\mathbb{F}_\ell$ has $\{g \cdot \varepsilon_n : g \bmod (x^m+1) \in M_{f_0}\} \subseteq RE_n^+$, i.e. the image of M_{f_0} in $A_n^{1/\ell}/A_n$ lies in the subgroup $RE_n^+/A_n = \{[\varepsilon] : \varepsilon \in RE_n^+ \cap A_n^{1/\ell}\}$ (a set of cosets). (KY Prop. 4.1, p.15, display (23).)*

Lemma 2.4 (KY trace floor [L]). *For $n \ge 3$ and $\varepsilon \in RE_n^+ \setminus \{\pm 1\}$: $\operatorname{Tr} \varepsilon^2 \ge 33 \cdot 2^n$ (KY Thm. 2.3, p.4). For $n = 2$: $\ge 17 \cdot 2^n$ (MO3 Prop. 6.6).*

Lemma 2.5 (Height floor [L]). *For $\varepsilon \in E_n \setminus E_{n-1}$ with $N_{n/n-1}\varepsilon = 1$: $\operatorname{ht}(\varepsilon) \ge L_n$. (MO16 Lemma 2.5(1), p.817.)*

Lemma 2.6 (Blichfeldt [L]). *A rank-d lattice Λ has a nonzero v with*

$$|v|^2 \le \frac{2}{\pi}\Gamma\big(1 + \tfrac{d+2}{2}\big)^{2/d} \operatorname{covol}(\Lambda)^{2/d}$$

(Bl14 Thm. II; MO16 Thm. 2.7). Equivalent power form: $|v|^d \le (2/\pi)^{d/2}\Gamma(2 + d/2) \operatorname{covol}(\Lambda)$.

Lemma 2.7 (Dirichlet [L]). *The logarithmic image of E_n in $\mathbb{R}^{2^n}$ is discrete; $L(1,\chi) \ne 0$ for nontrivial Dirichlet χ.*

Lemma 2.8 (Schinzel's inequality [L]). *Let ε be a totally real unit different from ± 1 and $\mathfrak{M}$ an ideal of $\mathbb{Q}(\varepsilon)$ containing $\varepsilon^2 - 1$, C the absolute norm of $\mathfrak{M}$, $d = [\mathbb{Q}(\varepsilon) : \mathbb{Q}]$. Then $M(\varepsilon) \ge \big((C^{1/d} + \sqrt{C^{2/d} + 4})/2\big)^{d/2}$. (MO3 Thm. 5.1, p. 1074, quoting Schinzel; verbatim in paper/sources/MO3_2020_clean.txt, line 631.)*

Lemma 2.9 (Height versus Mahler measure [L]). *For a unit ε of a totally real field of degree N: $\log M(\varepsilon) \le \frac{1}{2}\sqrt{N} \operatorname{ht}(\varepsilon)$ (MO16 eq. (2.4), proof of Lemma 2.3; Cauchy–Schwarz, reproved inside Lemma 6.3).*

Lemma 2.10 (Ramaré's bound for $|L(1,\chi)|$, even conductor [L]). *For every even primitive Dirichlet character χ of conductor q with $2 \mid q$ and $q \ge 4$: $|L(1,\chi)| \le U(q) := \frac{1}{4}(\log q + 2\log 2) + \log(4q)/\sqrt{q}$. (Ram04 Corollary 1 with $h = 1$, $k = 2$: $\omega(h) = 0$, $\phi(hk)/(2hk) = 1/4$, $\sum_{p|2} \log p/(p-1) = \log 2$, $\kappa_\chi = 0$ for even χ, error term $\phi(h)2^{\omega(k)-1}h^{-1}q^{-1/2}\log(q \cdot 4)$; condition $q \ge k^2 4^{\omega(h)} = 4$. Read from the author's accepted manuscript; the journal typesetting is to be compared before release.)*

Lemma 2.11 (Robbins' Stirling bound [L]). *For every integer $k \ge 1$: $k! = \sqrt{2\pi k}\,(k/e)^k e^{r_k}$ with $1/(12k+1) < r_k < 1/(12k)$; in particular $k! \le \sqrt{2\pi k}\,(k/e)^k e^{1/(12k)}$. (Rob55.)*

Lemma 2.12 (Horie's saturation lemma at odd p [L]; Ho05a Prop. 1 read in r17)**.** *p odd, $\ell \neq p$ prime, $F \subseteq \mathbb{Q}(\zeta_{p^n})$ an extension of the decomposition field of ℓ; η_n the Horie unit; $\alpha \mapsto \alpha_\sigma$ the coordinate lift $\mathbb{Z}[\zeta_{p^n}] \to \mathbb{Z}[\mathrm{Gal}]$. Then $\ell \mid h_{p,n}/h_{p,n-1}$ iff there is a prime $\mathcal{L}$ of F above ℓ with $\eta_n^{\alpha_\sigma}$ an ℓ-th power in $E_{p,n}$ for every $\alpha \in \ell\mathcal{L}^{-1}$. (MO13 Lemma 1.3, attributed to Ho05b, which relays Ho02 Lemmas 2, 3, 8; the same statement is quoted in Horie–Horie, Tohoku 61 (2009). Only the direction "$\ell \mid h_{p,n}/h_{p,n-1} \Rightarrow$" is used. r17: the general form of the lemma, with proof, is Ho05a Proposition 1 (open access; paper/Horie2005_JMSJ_triviality.pdf, read): for a Dirichlet character χ of order g_χ prime to ℓ, σ a generator of $\mathrm{Gal}(K_\chi/\mathbb{Q})$ and $k \subseteq \mathbb{Q}(\zeta_{g_\chi})$ containing the decomposition field of ℓ, $\ell \mid \bar{h}_\chi$ iff there is a prime $\mathfrak{l}$ of k above ℓ such that $|\eta_\chi^{\alpha_\sigma}|$ is an ℓ-th power in E_χ for every $\alpha \in \ell\mathfrak{l}^{-1}$; MO13 Lemma 1.3 is its specialisation to the layer ($K_\chi = B_{p,n}$, $\bar{h}_\chi = h_{p,n}/h_{p,n-1}$, η_χ the Horie unit up to sign and a root of unity). The original Ho02 (JLMS, paywalled) has still not been compared verbatim; Ho05a is by the same author and contains the proof.)*

Historical / superseded input (not a node of the graph; used by no current theorem). The r16 floor was the Schinzel-type Mahler-measure bound $M(\varepsilon) \geq ((1+\sqrt{5})/2)^{d/2}$ for a totally real unit $\varepsilon \neq \pm 1$ of degree d (MO13 Thm. 2.2 with the ideal $\mathfrak{M} = (1)$; Sch73); superseded in r17 by Lemma 7.2 and kept here only for the record (1.0.1, GPT r21 item 8).

Lemma 2.13 (Washington [L])**.** *For χ even primitive of conductor $f > 1$:*

$$L(1,\chi) = -\frac{\tau(\chi)}{f}\sum_{a=1}^{f} \bar{\chi}(a)\log|1-\zeta_f^a|, \qquad |\tau(\chi)| = \sqrt{f}.$$

(Wash, Chapter 4: the $L(1,\chi)$ formula for even primitive χ and the Gauss-sum modulus; cited at chapter level, hw 29.)

Chapter 3

The componentwise lemmas (Appendix A of the paper)

Lemma 3.1 (A: bridge [F]). *If $u \in RE_n^+ \cap E_{n-1}$ then $u = \pm 1$.*

Proof. Recall from the setting (notation section of the paper; Blueprint definition "setting) that $G_n = \langle \sigma \rangle$ is cyclic of order 2^n and $\tau = \sigma^{2^{n-1}}$ is its unique element of order 2; B_{n-1} is the fixed field of τ, so $\mathrm{Gal}(B_n/B_{n-1}) = \{1, \tau\}$ and the relative norm is $N_{n/n-1}(x) = x \cdot \tau(x)$ for $x \in B_n$. By definition $RE_n^+ = \{\varepsilon \in E_n : \ N_{n/n-1}\varepsilon = 1\}$, and $E_{n-1} = E_n \cap B_{n-1}$ is the set of units fixed by τ.

Let $u \in RE_n^+ \cap E_{n-1}$. Because $u \in E_{n-1}$ we have $\tau(u) = u$; because $u \in RE_n^+$ we have $u \cdot \tau(u) = 1$. Substituting the first relation into the second gives $u \cdot u = u^2 = 1$. (This is `bridge_fixed_norm_sq`, stated for an arbitrary abelian group G with a homomorphism $\tau : G \to G$: from $\tau u = u$ and $u \, \tau u = 1$ one obtains $u^2 = 1$; it is applied with $G = E_n$.)

In the field B_n the identity $u^2 = 1$ reads $(u-1)(u+1) = 0$, and a field has no zero divisors, so $u = 1$ or $u = -1$ (`torsion_pm_one`; only the field axioms of B_n are used, not that B_n is totally real). Hence $RE_n^+ \cap E_{n-1} \subseteq \{\pm 1\}$. Conversely ± 1 are units of B_{n-1} with $N_{n/n-1}(\pm 1) = (\pm 1)^2 = 1$, so in fact $RE_n^+ \cap E_{n-1} = \{\pm 1\}$; only the inclusion is used later. The Lean declaration(s) named above are this proof verbatim; `#print axioms` within std-3 (lean/README_lean.md). ☐

Lemma 3.2 (A^+: floor for relative units $\neq \pm 1$ [L-relative]). *$u \in RE_n^+ \setminus \{\pm 1\} \Rightarrow \mathrm{ht}(u) \geq L_n$.*

Proof. Let $u \in RE_n^+ \setminus \{\pm 1\}$. By Lemma 3.1, $u \notin E_{n-1}$, and $\mathrm{N}_{n/n-1}\, u = 1$ by the definition of RE_n^+. These are exactly the two hypotheses of [14] Lemma 2.5(1) (§3.3 of [14], p. 817: $\varepsilon \in E_n \setminus E_{n-1}$ with $\mathrm{Nr}_{B_n/B_{n-1}}(\varepsilon) = 1$), whose conclusion is $\mathrm{ht}(u) \geq \sqrt{2^n} \log(2 + \sqrt{5}) = L_n$. ☐

Lemma 3.3 (B: equal factor degrees [F]). *All irreducible factors of $x^m + 1$ over $\mathbb{F}_\ell$ have degree $d_n(\ell) = \mathrm{ord}_{2^n}(\ell)$. At $n = 7$: $d = 1$ for $\ell \equiv 1 \pmod{128}$ and $d = 2$ for $\ell \equiv 63, 65, 127 \pmod{128}$.*

Note. Kernel-checked (`WeberLemmaB.factor_degree`, `WeberLemmaB.factor_degree_eq`); the prose proof below is the same argument.

Proof. $x^m + 1 = \Phi_{2^n}(x)$ (the 2^n-th cyclotomic polynomial: $\Phi_{2^{k+1}}(x) = \Phi_2(x^{2^k}) = x^{2^k} + 1$); every root ζ has exact order 2^n, and $x^m + 1$ is separable over $\mathbb{F}_\ell$ because $\ell \nmid 2^n$. Over $\mathbb{F}_\ell$ (ℓ odd, unramified) the Frobenius $\zeta \mapsto \zeta^\ell$ permutes the roots, and the degree of the factor through ζ is the least k with $\zeta^{\ell^k} = \zeta$, i.e. $\ell^k \equiv 1 \pmod{2^n}$, the same for every root. (Lean: `WeberLemmaB.factor_degree` and `WeberLemmaB.factor_degree_eq`, std-3, using Mathlib's `Polynomial.natDegree_of_dvd_cyclotomic_of_irreducible` and `WeberLemmaB.X_pow_add_one_eq_cyclotomic`: $x^{2^{n-1}} + 1 = \Phi_{2^n}$.) At $n = 7$, $(\mathbb{Z}/128)^\times$ has exponent 32; $\ell \equiv 1$ has order 1; $63^2 = 3969 = 31 \cdot 128 + 1$, $65^2 = 4225 = 33 \cdot 128 + 1$, $127 \equiv -1$ have order 2. ☐

Lemma 3.4 (C: index of a component [F, quotient form]). *$[R_n : L_f] = \ell^{m-d_f}$, where $d_f = \deg f$.*

Note. $[R_n : L_f] = \ell^{m-d_f}$ is the dimension count of $(\mathbb{F}_\ell[x]/(x^m+1))/M_f$; the pullback identification along π is not formalised.

Proof. Since x^m+1 is separable mod ℓ, $\mathbb{F}_\ell[x]/(x^m+1) = \prod_g \mathbb{F}_\ell[x]/(g)$ over the irreducible factors. $w = (x^m+1)/f$ vanishes in every g-component with $g \neq f$ and is a unit in the f-component, so $M_f = 0 \times \cdots \times \mathbb{F}_\ell[x]/(f) \times \cdots \times 0$ has dimension d_f, and $R_n/L_f \cong (\mathbb{F}_\ell[x]/(x^m+1))/M_f$ has dimension $m - d_f$. The Lean statement `component_card_quotient` is the dimension count for the quotient V/W of $\mathbb{F}_\ell$-spaces; the pullback identification $R_n/L_f \cong V/W$ along π is the bookkeeping just described and is not formalised. □

Lemma 3.5 (D: rank and covolume [L-relative]). *H_n is injective, $H_n(\mathbb{Z}^m)$ is a discrete subgroup of $\mathbb{R}^{2m}$, hence a lattice of rank m with covolume $D_n = 2^{m/2}|\det W_n| > 0$, where $(W_n)_{j,i} = \lambda_{i+j}$ $(0 \le i, j < m)$. Consequently Λ_f is a lattice of rank m and covolume D_n/ℓ^{d_f}.*

Proof. (i) If $H_n a = 0$, every real embedding of u_a has absolute value 1; since B_n is totally real, every conjugate of u_a lies in $\{\pm 1\}$, so the minimal polynomial of u_a over $\mathbb{Q}$, being irreducible with all its roots in $\{\pm 1\}$, is $x-1$ or $x+1$, and $u_a = \pm 1$. (This is the totally real case of Kronecker's theorem—an algebraic integer all of whose conjugates have absolute value 1 is a root of unity; the general theorem is not needed.) freeness of $A_n/\{\pm 1\}$ ([6] §4) forces $a = 0$. (ii) $H_n(\mathbb{Z}^m)$ is the image of $A_n \subseteq E_n$ under the logarithmic embedding, whose image of E_n is discrete (Dirichlet); a subgroup of a discrete group is discrete. (iii) A discrete torsion-free subgroup of $\mathbb{R}^{2m}$ is free on $\mathbb{R}$-linearly independent generators; by (i) its $\mathbb{Z}$-rank is m, so its $\mathbb{R}$-rank is m. Antiperiodicity $\lambda_{j+m} = -\lambda_j$ gives $(H_n a)_{j+m} = -(H_n a)_j$, so $H_n^\top H_n = 2W_n^\top W_n$ and $D_n = \sqrt{\det H_n^\top H_n} = 2^{m/2}|\det W_n|$. For Λ_f: L_f has index ℓ^{m-d_f} in $\mathbb{Z}^m$ (Lemma C), so $H_n(L_f)$ has covolume $D_n \ell^{m-d_f}$, and scaling by ℓ^{-1} divides the covolume of a rank-m lattice by ℓ^m. □

Lemma 3.6 (E: saturation chain [L-relative; coset step F]). *Suppose $\ell \mid k_n$. Then there is an irreducible factor f_0 of x^m+1 over $\mathbb{F}_\ell$ such that for every $a \in L_{f_0}$ the real formal root r_a lies in RE_n^+, and its log vector is exactly $\ell^{-1}H_n a$.*

Note. The group step "a coset in RE_n^+/A_n has every representative in RE_n^+" is `coset_absorb` [F]; the rest consumes the [L] inputs.

Proof. Step 1 (the index). $\ell \mid k_n = [RE_n^+ : A_n]$ ([6] Eq. (17), an exact equality).

Step 2 (types). Write $[r_a] := r_a A_n$ for the coset of the real formal root $r_a \in A_n^{1/\ell}$ (a number) in the group $A_n^{1/\ell}/A_n$; the identification (1.1) sends $\bar{a} \in R_n/\ell R_n$ to $[r_a]$, and it is well defined because replacing a by $a + \ell b$ multiplies r_a by $u_b \in A_n$ (the real ℓ-th root is unique since ℓ is odd: `WeberRoots.odd_pow_injective`, `WeberRoots.root_change_rep`, `WeberRoots.coset_eq_of_change_rep`). Since $A_n \subseteq RE_n^+$ ([6] Def. 2.1), the set $\{[\varepsilon] : \varepsilon \in RE_n^+ \cap A_n^{1/\ell}\}$ is a subgroup of $A_n^{1/\ell}/A_n$; it is what [6] write RE_n^+/A_n.

Step 3 (the published input). [6] Prop. 4.1 supplies an irreducible f_0 with $\{g \cdot \varepsilon_n : g \in \mathbb{Z}[x],\ g \bmod (x^m+1) \in M_{f_0}\} \subseteq RE_n^+/A_n$, where $g \cdot \varepsilon_n$ is the real root r_g and the set is a set of cosets; in the identification (1.1) this says that every coset $[r_a]$ with $\bar{a} \in M_{f_0}$, i.e. with $a \in L_{f_0}$, belongs to the subgroup of Step 2: $[r_a] = [\varepsilon]$ for some $\varepsilon \in RE_n^+ \cap A_n^{1/\ell}$. (We do not say that a coset is a subset of RE_n^+/A_n; membership of the coset in the image of $RE_n^+ \cap A_n^{1/\ell}$ is the statement.)

Step 4 (from the coset to the number). $[r_a] = [\varepsilon]$ means $r_a = \varepsilon c$ with $c \in A_n$. Since $A_n \subseteq RE_n^+$ and RE_n^+ is a group, $r_a \in RE_n^+$. This holds for every $a \in L_{f_0}$; if a is replaced by $a + \ell b$, r_a is multiplied by $u_b \in A_n \subseteq RE_n^+$, so membership in RE_n^+ is preserved, as it must be since $\ell R_n \subseteq L_{f_0}$.

Step 5 (the log vector). r_a is the real ℓ-th root of u_a, so $\ell \log|\sigma^j r_a| = \log|\sigma^j u_a| = (H_n a)_j$ for every real embedding σ^j, $j < 2m$: the log vector of r_a is $\ell^{-1}H_n a$. The base branch $a \in \ell R_n$ is Lemma E′. ☐

Lemma 3.7 (E′: the base case $a \in \ell R_n$ [L-relative]). *For $a = \ell b$, $b \in R_n$, the vector $\ell^{-1}H_n a = H_n b$ is the log vector of the unit $u_b \in A_n \subseteq RE_n^+$, and $u_b = \pm 1$ if and only if $b = 0$.*

Proof. The identity $\ell^{-1}H_n(\ell b) = H_n b$ is linear algebra; $u_b \in A_n \subseteq RE_n^+$ by [6] Def. 2.1; and $u_b = \pm 1$ forces $b = 0$ by the freeness of $A_n/\{\pm 1\}$ ([6] §4). ☐

Theorem 3.8 (SH: componentwise saturation–height bound, carrier form [M; F-core]). *Let K be a totally real number field of degree N, $H : K^\times \to \mathbb{R}^N$ the logarithmic map over its real embeddings,* $\mathrm{ht} = |H(\cdot)|$, *and $U \subseteq \mathcal{O}_K^\times$ any set of units. Let R be a free abelian group of rank m and $h : R \to \mathbb{R}^N$ an additive map with (Rank): h is injective; let D be the covolume of the lattice $h(R)$ in its $\mathbb{R}$-span and $K_m = (2/\pi)^{m/2}\Gamma(2 + m/2)$. Let ℓ be an odd prime and $L \subseteq R$ a subgroup with (Index): $\ell R \subseteq L$ and $[R : L] = \ell^{m-d}$; let $L_0 > 0$. Assume*

- *(Car) for every $a \in L$ there is $u \in U$ with $\ell H(u) = h(a)$ (the scaled lattice vector $\ell^{-1}h(a)$ is carried by a unit of U);*
- *(Floor) every $u \in U$ with $H(u) \neq 0$ has $\mathrm{ht}(u) \geq L_0$;*
- *(Bl) every lattice Λ of rank m in $\mathbb{R}^N$ contains $v \neq 0$ with $|v|^m \leq K_m \operatorname{covol}(\Lambda)$ ([1] Thm. II, as [14] Thm. 2.7).*

Then $\ell^d \leq K_m D / L_0^m$.

Proof. Covolume. Put $\Lambda := \ell^{-1}h(L) = \{\ell^{-1}h(a) : a \in L\}$. By (Rank) h is an injective $\mathbb{Z}$-linear map $R \to \mathbb{R}^N$, so $h(L)$ is a sublattice of $h(R)$ of index $[R : L] = \ell^{m-d}$ (Index) and has covolume $\ell^{m-d}D$ in the common m-dimensional $\mathbb{R}$-span; the map $v \mapsto \ell^{-1}v$ on that span multiplies m-dimensional volumes by ℓ^{-m}, so $\operatorname{covol}\Lambda = \ell^{m-d}D/\ell^m = D/\ell^d$.

Short vector. By (Bl) there is $a \in L$ with $v := \ell^{-1}h(a) \neq 0$ and $|v|^m \leq K_m \operatorname{covol}\Lambda = K_m D/\ell^d$.

Carrier and floor. By (Car) there is $u \in U$ with $\ell H(u) = h(a)$, i.e. $H(u) = v \neq 0$, and (Floor) gives $|v| = \mathrm{ht}(u) \geq L_0$ (`WeberSH.short_vector_ge_floor`).

Hence $L_0^m \leq |v|^m \leq K_m D/\ell^d$, i.e. $\ell^d \leq K_m D/L_0^m$ (`WeberSH.theoremSH`; contrapositive `WeberSH.theoremSH_contra`). The covolume computation is the natural-language part of this proof; the short-vector, carrier and floor steps are the Lean theorem verbatim, with $h(a) = H(\iota(a))$ for the map $\iota : \mathbb{Z}^m \to U$ of the Lean statement. Nothing about R beyond additivity of h, nothing about U beyond (Car) and (Floor), and no group action is used: these enter only through how L, d and (Car) are produced in applications (Corollary 3.9 for the $\mathbb{Z}_2$-tower; Lemma 7.3 for the $\mathbb{Z}_3$-tower). ☐

Corollary 3.9 (SH-mod: the module form [M; F-core]). *In the setting of Theorem 3.8 let moreover G be a finite abelian group acting on K by automorphisms, R a commutative quotient ring of $\mathbb{Z}[G]$ (free of rank m over $\mathbb{Z}$), U a G-stable subgroup of $\mathcal{O}_K^\times$, $A \subseteq U$ a G-stable subgroup containing ± 1 such that $A/\{\pm 1\}$ is a cyclic R-module generated by the class of $\varepsilon \in A$, $u_a = \varepsilon^a$ ($a \in R$), $h(a) = H(u_a)$, and $L = \pi^{-1}(M)$ the pullback of a nonzero ideal M of dimension d of the $\mathbb{F}_\ell$-algebra $R/\ell R$. If*

- *(Sat) for every $a \in L \setminus \ell R$ there is $u \in U$ with $u^\ell = u_a$,*

then (Car) holds, and with (Floor) and (Bl) as in Theorem 3.8, $\ell^d \leq K_m D/L_0^m$.

Proof. $h(a) = H(u_a)$ is additive in a (`WeberSH.H_pow`: $H(u^k) = kH(u)$, and $u_{a+b} = \pm u_a u_b$ is invisible to H), so Theorem 3.8 applies once (Car) is verified for every $a \in L$.

Base branch ($a \in \ell R$). Write $a = \ell b$ with $b \in R$. Then $u_b \in A \subseteq U$ and $u_b^\ell = \pm u_a$, so $\ell H(u_b) = H(u_a) = h(a)$ (`WeberSH.base_branch`).

Saturation branch ($a \in L \setminus \ell R$). By (Sat) there is $u \in U$ with $u^\ell = u_a$; then $\ell H(u) = H(u_a) = h(a)$ (`WeberSH.sat_root_log`).

In both branches the carrier exists; Theorem 3.8 gives $\ell^d \le K_m D/L_0^m$. (For a unit u of a totally real field, $H(u) = 0$ iff all conjugates are ± 1 iff $u = \pm 1$; so (Floor) in the form "$u \ne \pm 1 \Rightarrow \mathrm{ht}(u) \ge L_0$" is the same hypothesis.) □

Proposition 3.10 (F: factorwise bound [L-relative])**.** *If $\ell \mid k_n$ and f_0 is the saturation component of Lemma 3.6, then $\ell^{d_{f_0}} \le C_n$.*

Proof. Apply Corollary 3.9 (the module form of Theorem 3.8) with $K = B_n$, $N = 2^n$, $G = G_n$, $R = R_n = \mathbb{Z}[x]/(x^m + 1)$ (the quotient of $\mathbb{Z}[G_n]$ by $1 + \tau$, which acts on A_n as the norm to B_{n-1}, trivial on $RE_n^+ \supseteq A_n$), $U = RE_n^+$, $A = A_n$, $\varepsilon = \varepsilon_n$ (so u_a is the unit $\prod_i \sigma^i(\varepsilon_n)^{a_i}$ of the setting; $A_n/\{\pm 1\}$ is free on $\sigma^i \varepsilon_n$, [6] §4, hence cyclic over R_n), (Rank) by Lemma 3.5, $D = D_n$, $M = M_{f_0}$ of dimension $d_{f_0} = \deg f_0$, $L = L_{f_0}$ with (Index) by Lemma 3.4, and $L_0 = L_n$. Hypothesis (Sat) is Lemma 3.6: for $a \in L_{f_0} \setminus \ell R_n$ the real root $r_a \in RE_n^+$ satisfies $r_a^\ell = u_a$ (the base branch $a \in \ell R_n$ of Corollary 3.9 is Lemma 3.7). Hypothesis (Floor) is Lemma 3.2 (Lemma 3.1 with [14] Lemma 2.5(1)). Hypothesis (Bl) is [1] Thm. II in the form [14] Thm. 2.7. Corollary 3.9 gives $\ell^{d_{f_0}} \le K_m D_n / L_n^m = C_n$.

Unfolded, for the reader who wants the argument in place: $\Lambda_{f_0} = \ell^{-1} H_n(L_{f_0})$ has rank m and covolume D_n/ℓ^{d_0} (Lemma 3.5); if $\ell^{d_0} > C_n$, Blichfeldt gives $0 \ne v = \ell^{-1} H_n a \in \Lambda_{f_0}$ with $|v|^m \le (2/\pi)^{m/2} \Gamma(2 + m/2) D_n/\ell^{d_0} \le C_n L_n^m/\ell^{d_0} < L_n^m$; if $a \in \ell R_n$, Lemma 3.7 makes v the log vector of $u_b \in RE_n^+ \setminus \{\pm 1\}$, if $a \notin \ell R_n$, Lemma 3.6 makes v the log vector of $r_a \in RE_n^+ \setminus \{\pm 1\}$; Lemma 3.2 gives $\mathrm{ht} \ge L_n$, contradicting $|v| < L_n$. □

Chapter 4

Old and new components

Lemma 4.1 (old/new decomposition [M]). *Let L/K be a cyclic extension of number fields of degree p^k and $\ell \neq p$ a prime. Let $\iota : \mathrm{Cl}(K) \to \mathrm{Cl}(L)$ be extension of ideals and $N : \mathrm{Cl}(L) \to \mathrm{Cl}(K)$ the norm. Then $\mathrm{Cl}(L)_\ell = \iota(\mathrm{Cl}(K)_\ell) \oplus \ker(N)_\ell$, ι is injective on $\mathrm{Cl}(K)_\ell$, and $v_\ell(h_L) > v_\ell(h_K)$ if and only if $\ker(N)$ has an element of order ℓ. (In the Weber tower $k_n = h_n/h_{n-1}$ is an integer by the index formula of [6], and $v_\ell(h_n) > v_\ell(h_{n-1})$ is then $\ell \mid k_n$.)*

Proof. Throughout, $\mathrm{Cl}(\cdot)$ is the ideal class group (finite abelian), $\mathrm{Cl}(\cdot)_\ell$ its ℓ-primary part (the subgroup of elements of ℓ-power order), $\iota : \mathrm{Cl}(K) \to \mathrm{Cl}(L)$ the map induced by extension of ideals $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}_L$, and $N : \mathrm{Cl}(L) \to \mathrm{Cl}(K)$ the map induced by the ideal norm $N_{L/K}$. Both are group homomorphisms; both restrict to the ℓ-parts because a homomorphism of finite abelian groups maps elements of ℓ-power order to elements of ℓ-power order.

Step 1: $N \circ \iota$ is multiplication by $[L : K]$. For an ideal $\mathfrak{a}$ of K, $N_{L/K}(\mathfrak{a}\mathcal{O}_L) = \mathfrak{a}^{[L:K]}$ (transitivity of the norm: the norm of an extended ideal is the $[L : K]$-th power). Hence $N \circ \iota = [L : K] = p^k$ on $\mathrm{Cl}(K)$.

Step 2: injectivity on the ℓ-part. Multiplication by p^k is a bijection of the finite abelian ℓ-group $\mathrm{Cl}(K)_\ell$ because $\gcd(p^k, |\mathrm{Cl}(K)_\ell|) = 1$ ($\ell \neq p$); an inverse is multiplication by an integer u with $up^k \equiv 1$ modulo the exponent of $\mathrm{Cl}(K)_\ell$. Since $N \circ \iota$ is bijective on $\mathrm{Cl}(K)_\ell$, ι is injective there, and $\iota(\mathrm{Cl}(K)_\ell) \cap \ker N = 0$: an element ιy with $y \in \mathrm{Cl}(K)_\ell$ and $N\iota y = 0$ has $p^k y = 0$, hence $y = 0$.

Step 3: the decomposition. Let $x \in \mathrm{Cl}(L)_\ell$. Then $N(x) \in \mathrm{Cl}(K)_\ell$, and we may put $y := u\,N(x)$ with u as in Step 2, so that $p^k y = N(x)$. Then $N(x-\iota y) = N(x) - N\iota y = N(x) - p^k y = 0$, i.e. $x - \iota y \in \ker N$; it lies in $\mathrm{Cl}(L)_\ell$ as a difference of two elements of ℓ-power order. Thus $x = \iota y + (x - \iota y)$ exhibits $\mathrm{Cl}(L)_\ell = \iota(\mathrm{Cl}(K)_\ell) + \ker(N)_\ell$, and the sum is direct by Step 2. Consequently

$$|\mathrm{Cl}(L)_\ell| = |\mathrm{Cl}(K)_\ell| \cdot |\ker(N)_\ell|.$$

Step 4: the valuation statement. For a finite abelian group G, $|G_\ell| = \ell^{v_\ell(|G|)}$, so $v_\ell(h_L)$ equals v_ℓ of $|\mathrm{Cl}(L)_\ell|$, and likewise for K. Taking v_ℓ of the displayed identity, $v_\ell(h_L) - v_\ell(h_K) = v_\ell(|\ker(N)_\ell|) \geq 0$, with equality iff $\ker(N)_\ell = 0$. Hence $v_\ell(h_L) > v_\ell(h_K)$ iff $\ker(N)_\ell \neq 0$.

Step 5: the Weber tower. Apply this with $K = B_{n-1}$, $L = B_n$, $p = 2$, $k = 1$: B_n/B_{n-1} is cyclic of degree 2 (Galois group $\langle \tau \rangle$). Then $v_\ell(h_n) \geq v_\ell(h_{n-1})$ for every odd ℓ, which is the odd part of the integrality $h_{n-1} \mid h_n$ ([6] Eq. (17): $k_n = h_n/h_{n-1} \in \mathbb{Z}$ and $k_n = [RE_n^+ : A_n]$), and

$$\ell \mid k_n \iff v_\ell(h_n) > v_\ell(h_{n-1}) \iff \ker(N)_\ell \neq 0 :$$

the odd primes dividing k_n are exactly the odd primes of the "new" part $\ker N$ of the class group. This is the sense in which k_n, and not h_n, is the natural object at a fixed layer; [6] Eq. (17) is the description of k_n by relative units that Theorem A uses. Nothing about $\ell = p = 2$ is claimed. □

Chapter 5

Main results

Theorem 5.1 (A core: discrete implication [F])**.** *Given a carrier of components with predicates* saturated*, a height* ht*, constants* $L, K, D, \ell^d > 0$*,* $m > 0$*, and the hypotheses (ky41)* $\ell \mid k_n \Rightarrow \exists f$ *saturated; (blichfeldt)* $\forall f$*,* $\mathrm{ht}(f)^m \le KD/\ell^d$*; (floor) saturated* $f \Rightarrow L \le \mathrm{ht}(f)$*: if* $C := KD/L^m < \ell^d$ *then* $\ell \nmid k_n$*.*

Proof. We write the argument so that it can be followed without the Lean source. There are three data: a set of "components" f with a predicate *saturated*(f) and a real number $\mathrm{ht}(f)$ attached to each f; positive real constants L (a height floor), K (the Blichfeldt constant), D (a covolume) and ℓ^d (the index of the component lattice), and an integer $m > 0$ (the rank); and three hypotheses, each a plain implication or inequality with no quantifier hidden inside a definition:

(ky41) if $\ell \mid k_n$, then some component f is saturated;

(blichfeldt) for every component f, $\mathrm{ht}(f)^m \le KD/\ell^d$;

(floor) for every saturated component f, $L \le \mathrm{ht}(f)$.

Put $C := KD/L^m$ and assume $C < \ell^d$. We prove $\ell \nmid k_n$.

Suppose, for a contradiction, that $\ell \mid k_n$. By (ky41) there is a saturated component f. By (floor), $L \le \mathrm{ht}(f)$. Since $0 < L$ and m is a positive integer, the function $y \mapsto y^m$ is increasing on $y \ge 0$, so $L^m \le \mathrm{ht}(f)^m$. By (blichfeldt), $\mathrm{ht}(f)^m \le KD/\ell^d$. Chaining the two inequalities, $L^m \le KD/\ell^d$.

On the other hand the assumption $C < \ell^d$ reads $KD/L^m < \ell^d$. Multiply both sides by $L^m > 0$: $KD < \ell^d L^m$. Divide both sides by $\ell^d > 0$: $KD/\ell^d < L^m$. The two displayed conclusions, $L^m \le KD/\ell^d$ and $KD/\ell^d < L^m$, contradict each other. Hence $\ell \nmid k_n$.

In the paper the three hypotheses are instantiated as follows (this instantiation is Prop. F, not this theorem): components are the irreducible factors f of $x^m + 1$ over $\mathbb{F}_\ell$; *saturated* is the conclusion of [6] Prop. 4.1 for the component f_0 of Lemma 3.6; $\mathrm{ht}(f)$ is the length of the shortest nonzero vector of Λ_f; $L = L_n$ is the floor of Lemma A$^+$; $K = (2/\pi)^{m/2}\Gamma(2 + m/2)$ and $D = D_n$ are the constants of Blichfeldt's theorem and of Lemma D, so that (blichfeldt) is the power-$m/2$ form of Blichfeldt's inequality for the lattice Λ_f of covolume D_n/ℓ^d; and $C = C_n$. In that instantiation (floor) is where the saturation description enters (a short vector of Λ_{f_0} is the log vector of a relative unit $\ne \pm 1$, whose height is at least L_n), and (ky41) is the existence of the saturation component.

Lean. `TheoremAInputs.theoremA_of_inputs`: the three hypotheses are the fields `ky41`, `blichfeldt`, `floor` of the structure `TheoremAInputs`; the two rewritings of $C < \ell^d$ are `div_lt_iff`; the final contradiction is `linarith`. Axiom census std-3 (`lean/WeberExternalResults.lean`). The Lean declaration(s) named above are this proof verbatim; `#print axioms` within std-3 (lean/README_lean.md). □

Theorem 5.2 (A [L-relative, F-core]). *Let $n \geq 2$, ℓ an odd prime and $d_n(\ell) = \mathrm{ord}_{2^n}(\ell)$. Let D_n be the covolume of the log-lattice $H_n(\mathbb{Z}^m)$ of A_n (Definition 1.1) and*

$$C_n = (2/\pi)^{m/2}\,\Gamma(2+m/2)\,D_n/L_n^m.$$

If $\ell^{d_n(\ell)} > C_n$ then $\ell \nmid k_n$.

Proof. Suppose $\ell \mid k_n$. Proposition 3.10 gives $\ell^{d_{f_0}} \leq C_n$ for the saturation component f_0, and Lemma 3.3 gives $d_{f_0} = d_n(\ell)$; hence $\ell^{d_n(\ell)} \leq C_n$. Contrapositively, $\ell^{d_n(\ell)} > C_n$ implies $\ell \nmid k_n$. The discrete core of this argument (saturation exists, every lattice vector obeys the Blichfeldt bound in the power-$m/2$ form, saturation forces height $\geq L$, hence $C < \ell^d$ is impossible) is the Lean theorem `TheoremAInputs.theoremA_of_inputs`. ☐

Lemma 5.3 (D0: group determinant). $D_n = 2^{-m/2}\prod_{\chi(\tau)=-1}|S(\chi)|$, *where* $S(\chi) = \sum_{g\in G_n}\chi(g)\log|g\varepsilon_n|$.

Note. Paper Appendix D, Lemma 0; second route to the normalisation: ball-arithmetic eigenvector check, `sage/r11_propD_audit.log`.

Proof. $D_n^2 = \det(H_n^\top H_n)$, and

$$(H_n^\top H_n)_{ik} = \sum_{j<2m}\lambda_{i+j}\lambda_{k+j} = c(k-i), \qquad c(t) = \sum_{j \bmod 2m}\lambda_j\lambda_{j+t},$$

an even function with $c(t+m) = -c(t)$. For $\omega^m = -1$ and $v = (\omega^i)_{i<m}$, $(H_n^\top H_n v)_i = \omega^i\mu(\omega)$ with $\mu(\omega) = \sum_{u<m} c(u)\omega^{-u} = \frac{1}{2}\sum_{u \bmod 2m} c(u)\omega^{-u} = \frac{1}{2}|\hat{\lambda}(\omega)|^2$, $\hat{\lambda}(\omega) = \sum_{j<2m}\lambda_j\omega^j$, using the m-periodicity of $c(u)\omega^{-u}$. The m vectors v for the m roots of x^m+1 are independent (Vandermonde), so $\det(H_n^\top H_n) = 2^{-m}\prod_\omega|\hat{\lambda}(\omega)|^2$; with $\omega = \chi(3)$, $\hat{\lambda}(\omega) = S(\chi)$. This gives Prop. D(i). An independent check of the normalisation $2^{-m/2}$: the eigenvector identity $H_n^\top H_n v = \frac{1}{2}|S(\chi)|^2 v$ was verified in ball arithmetic for $n = 2,\ldots,7$ with residual $< 10^{-143}$, and $\sqrt{\det H_n^\top H_n}$ agrees with $2^{-m/2}\prod|S(\chi)|$ to the precision of the determinant ball (`sage/r11_propD_audit.log`). ☐

Lemma 5.4 (D1: cyclotomic-unit expression). $\varepsilon_n = (1-\zeta^3)^2(1-\zeta^2)/((1-\zeta)^2(1-\zeta^6))$ *with* $\zeta = e^{2\pi i/q}$, $q = 2^{n+2}$.

Proof. With $\zeta = e^{2\pi i/q}$ and $X = \zeta+\zeta^{-1}$: $X+1 = \zeta^{-1}(\zeta^3-1)/(\zeta-1)$, $X-1 = \zeta^{-1}(\zeta^3+1)/(\zeta+1)$, hence $\varepsilon_n = (1-\zeta^3)^2(1-\zeta^2)/((1-\zeta)^2(1-\zeta^6))$; all factors are nonzero for $q \geq 16$. Consequently $\log|\sigma_a\varepsilon_n| = 2\log|1-\zeta^{3a}| - 2\log|1-\zeta^a| + \log|1-\zeta^{2a}| - \log|1-\zeta^{6a}|$ for $a \in (\mathbb{Z}/q)^\times$. ☐

Lemma 5.5 (D2: imprimitive terms vanish). *For χ of conductor exactly q and $c \in \{2,6\}$:* $\sum_{a\in(\mathbb{Z}/q)^\times}\chi(a)\log|1-\zeta^{ca}| = 0$.

Proof. For χ of conductor exactly q and $c \in \{2,6\}$, $\sum_{a\in(\mathbb{Z}/q)^\times}\chi(a)\log|1-\zeta^{ca}| = 0$: ζ^{ca} depends only on $a \bmod q/2$, so the summand is constant on cosets of $U = \{1, 1+q/2\}$, and $\sum_{u\in U}\chi(u) = 1+\chi(1+q/2) = 0$. ☐

Lemma 5.6 (D3: Washington's formula in the form used [L]). *For χ even and primitive of conductor $f > 1$:* $\left|\sum_{a\in(\mathbb{Z}/f)^\times}\chi(a)\log|1-\zeta_f^a|\right| = \sqrt{f}\,|L(1,\chi)|$.

Proof. For χ even and primitive of conductor $f > 1$ ([20], Chapter 4), and with $|\tau(\chi)| = \sqrt{f}$,

$$L(1,\chi) = -\frac{\tau(\chi)}{f}\sum_{a=1}^{f}\bar{\chi}(a)\log|1-\zeta_f^a|.$$

Applying this to $\bar{\chi}$ and using $|L(1,\bar{\chi})| = |L(1,\chi)|$:

$$\Big|\sum_{a\in(\mathbb{Z}/f)^\times} \chi(a)\log|1-\zeta_f^a|\Big| = \sqrt{f}\,|L(1,\chi)|.$$

The conjugation convention does not affect the absolute value. □

Lemma 5.7 (D4: $\prod|1-\chi(3)| = 2$ for every n [M])**.** *For every $n \geq 2$: $\prod_{\chi(\tau)=-1}|1-\chi(3)| = 2$.*

Proof. By Appendix C (Frobenius orbits), as χ runs over the characters of G_n with $\chi(\tau) = -1$, i.e. the even characters of conductor exactly $q = 2^{n+2}$, the value $\chi(3)$ runs exactly once over the primitive 2^n-th roots of unity ω: indeed χ is determined by $\chi(3)$ because 3 generates G_n, and $\chi(3)^m = \chi(\tau) = -1$ with $m = 2^{n-1}$ says that $\chi(3)$ is a root of $x^m+1 = \Phi_{2^n}(x)$, all of whose m roots are primitive 2^n-th roots of unity. Hence $\prod_\chi(1-\chi(3)) = \prod_{\omega\text{ primitive}}(1-\omega) = \Phi_{2^n}(1) = 1^m+1 = 2$, and taking absolute values, $\prod_\chi|1-\chi(3)| = 2$ for every $n \geq 2$. (The symbolic check in $\mathbb{Q}(\zeta_{2^n})$ for $n \leq 12$ in `sage/r11_propD_audit.log` is a numerical confirmation of this identity, not its proof.) □

Proposition 5.8 (D: evaluation of D_n [L-relative; classical])**.** *With $S(\chi) = \sum_{g\in G_n}\chi(g)\log|g\varepsilon_n|$ and χ ranging over the m characters of G_n with $\chi(\tau) = -1$, equivalently the even Dirichlet characters of conductor exactly q:*

$$D_n = 2^{-m/2}\prod_\chi|S(\chi)| \qquad \text{and} \qquad D_n = 2^{m(n+1)/2+1}\prod_\chi|L(1,\chi)|.$$

Note. Cross-CAS: r10_bin4 (PARI/Magma), r11_propD_audit (Arb digamma route).

Proof. (i) is Lemma 5.3. (ii): Each $g \in G_n$ has two lifts $\pm a$ with the same $\log|\sigma_a\varepsilon_n|$, so $S(\chi) = \frac{1}{2}\sum_{a\in(\mathbb{Z}/q)^\times}\chi(a)\log|\sigma_a\varepsilon_n|$ (this half-sum normalisation was checked independently in ball arithmetic, $n = 2,\ldots,7$, residual $< 10^{-143}$). Insert Lemma 5.4 and drop the $c = 2, 6$ terms by Lemma 5.5: $S(\chi) = \sum_a\chi(a)\log|1-\zeta^{3a}| - \sum_a\chi(a)\log|1-\zeta^a| = (\bar{\chi}(3)-1)\sum_a\chi(a)\log|1-\zeta^a|$ by the substitution $b = 3a$. Lemma 5.6 gives $|S(\chi)| = |1-\chi(3)|\sqrt{q}\,|L(1,\chi)|$. Then Lemma 5.3 with $\prod_\chi|1-\chi(3)| = 2$ (Lemma 5.7, every $n \geq 2$) and $q^{m/2} = 2^{(n+2)m/2}$: $D_n = 2^{-m/2+(n+2)m/2+1}\prod_\chi|L(1,\chi)| = 2^{m(n+1)/2+1}\prod_\chi|L(1,\chi)|$. □

Corollary 5.9 (A′ [follows])**.** *$C_n = 2\,(4/\pi)^{m/2}\,\Gamma(2+m/2)\prod_\chi|L(1,\chi)|/\log(2+\sqrt{5})^m$.*

Proof. $L_n^m = 2^{nm/2}\log(2+\sqrt{5})^m$, and $2^{m(n+1)/2+1}/2^{nm/2} = 2^{m/2+1}$, $(2/\pi)^{m/2}2^{m/2} = (4/\pi)^{m/2}$. □

Corollary 5.10 (T: order threshold [F])**.** *For $B > 1$ put $f_n(B) = \max\{1,\,1+\lfloor\log C_n/\log B\rfloor\}$. If $\ell \geq B$ and $d_n(\ell) \geq f_n(B)$ then $\ell \nmid k_n$.*

Note. Read relative to the core hypotheses (the [L] inputs), as Theorem A.

Proof. Let $B > 1$, $\ell \geq B$ and $d = d_n(\ell) \geq f_n(B)$, where $f_n(B) = \max\{1,\,1+\lfloor\log C_n/\log B\rfloor\}$ and $C_n > 0$ (Prop. D: $D_n > 0$).

The integer floor. Write $x = \log C_n/\log B \in \mathbb{R}$ (the sign of x is irrelevant; for $C_n < 1$ it is negative). By definition of the floor, $\lfloor x\rfloor \leq x < \lfloor x\rfloor + 1$, so $f_n(B) \geq \lfloor x\rfloor + 1 > x$; this is the only property of $f_n(B)$ used, and it is why the "+1" is in the definition: $\lfloor x\rfloor + 1$ is the least integer strictly larger than x. The outer $\max\{1,\cdot\}$ only matters when $C_n < 1$ (then $x < 0$ and $\lfloor x\rfloor + 1 \leq 0$); it keeps $f_n(B) \geq 1 = \min_\ell d_n(\ell)$, so that the hypothesis $d \geq f_n(B)$ is never vacuous, and it does not affect the inequality $f_n(B) > x$. (Without the max the statement would still be true, because $d \geq 1$ for every odd ℓ; the max removes the case distinction.)

From the floor to the constant. Since $B > 1$, $\log B > 0$ and $x \mapsto B^x = \exp(x \log B)$ is strictly increasing on $\mathbb{R}$. Hence $B^{f_n(B)} > B^x = \exp(\log C_n) = C_n$. If $d \geq f_n(B)$ then $B^d \geq B^{f_n(B)}$ (monotonicity again, $B > 1$), so $B^d > C_n$. If instead $d < f_n(B)$ is excluded by hypothesis, nothing else is needed: we only use $d \geq f_n(B)$ and $d \geq 1$.

From B to ℓ. $\ell \geq B > 1$ and $d \geq 1$ give $\ell^d \geq B^d$ (monotonicity of $y \mapsto y^d$ on $y > 0$ for a positive integer exponent). Therefore $\ell^{d_n(\ell)} = \ell^d \geq B^d > C_n$, which is the hypothesis $\ell^{d_n(\ell)} > C_n$ of Theorem 5.2, and Theorem 5.2 gives $\ell \nmid k_n$.

Example (the numbers of Cor. 7). At $n = 7$ with $B = 1314283897427173$: $\log C_7 / \log B = \log_{10} C_7 / \log_{10} B = 30.2374\ldots/15.1187\ldots = 1.99998\ldots$ (the ratio does not depend on the base; in natural logarithms $69.6241\ldots/34.8121\ldots$), so $f_7(B) = 1 + 1 = 2$; every prime $\ell \geq B$ with $d_7(\ell) \geq 2$, i.e. $\ell \not\equiv 1 \pmod{128}$, satisfies $\ell^2 \geq B^2 > C_7$ (the integer form of this comparison is Lemma 5.14). With $B = 1727342163036353095979941756930$ the quotient is $0.99999\ldots$, $f_7(B) = 1$, and every prime $\ell \geq B$ is covered whatever its order. Both thresholds are the frozen integers of Cor. 5.15.

Lean. `TheoremAInputs.theoremA_threshold` takes the two comparisons $C < B^d$ and $B^d \leq \ell^d$ as hypotheses and feeds them to the core `theoremA_of_inputs`; the floor computation above is the natural-language justification of those two hypotheses and is not itself formalised (it is ordinary real-number monotonicity). □

Corollary 5.11 ($\widehat{\mathrm{A}}$: computation-free criterion [M relative to Theorem A's inputs + L: Ram04]). *For even $q \geq 4$ put $U(q) = \frac{1}{4}(\log q + 2\log 2) + \log(4q)/\sqrt{q}$, and*

$$\widehat{C}_n = 2\,(4/\pi)^{m/2}\,\Gamma(2 + m/2)\,\bigl(U(2^{n+2})/\log(2+\sqrt{5})\bigr)^m.$$

Then $C_n \leq \widehat{C}_n$ for every $n \geq 2$, and for every odd prime ℓ: $\ell^{d_n(\ell)} > \widehat{C}_n$ implies $\ell \nmid k_n$.

Proof. By Corollary 5.9,

$$C_n = 2\,(4/\pi)^{m/2}\,\Gamma(2+m/2) \prod_{\chi} |L(1,\chi)| / \log(2+\sqrt{5})^m,$$

the product over the m even Dirichlet characters of conductor exactly $q = 2^{n+2}$ (Proposition 5.8). Each such χ is primitive and even, $2 \mid q$ and $q \geq 16 \geq 4$, so [16] Corollary 1 with $h = 1$ and $k = 2$ gives $0 \leq |L(1,\chi)| \leq U(q)$. A product of m real numbers in $[0, U(q)]$ is at most $U(q)^m$; hence

$$C_n \leq 2\,(4/\pi)^{m/2}\,\Gamma(2+m/2)\,U(q)^m / \log(2+\sqrt{5})^m = \widehat{C}_n.$$

If $\ell^{d_n(\ell)} > \widehat{C}_n$ then $\ell^{d_n(\ell)} > C_n$, and Theorem 5.2 gives $\ell \nmid k_n$. The discrete part (a product of m factors in $[0, U]$ is $\leq U^m$, and $C \leq \widehat{C} < \ell^d$ feeds $C < \ell^d$ into the discrete core `TheoremAInputs.theoremA_of_inputs`) is the Lean theorems `WeberHatC.prod_le_pow_card_of_le`, `WeberHatC.C_le_hatC` and `WeberHatC.theoremA_of_hat`; the analytic input $|L(1,\chi)| \leq U(q)$ is a literature hypothesis and is not formalised. □

Corollary 5.12 (explicit order threshold [M relative to cor:Ahat + L: Rob55]). *Put $k = 1 + m/2$ and*

$$T_n = \log 2 + \tfrac{m}{2}\log\tfrac{4}{\pi} + \tfrac{1}{2}\log(2\pi k) + k(\log k - 1) + \tfrac{1}{12k} + m\log\frac{U(2^{n+2})}{\log(2+\sqrt{5})}.$$

Then $\log \widehat{C}_n \leq T_n$, and for every odd prime ℓ: $d_n(\ell)\log\ell > T_n$ implies $\ell \nmid k_n$.

Proof. Since $n \geq 2$, $m/2 = 2^{n-2}$ is an integer and $\Gamma(2 + m/2) = (1 + m/2)! = k!$ with $k = 1 + m/2 \geq 2$ (for a positive integer k, $\Gamma(k+1) = k!$). Robbins' form of Stirling's formula ([17])

gives $k! \le \sqrt{2\pi k}\,(k/e)^k e^{1/(12k)}$, i.e. $\log k! \le \frac{1}{2}\log(2\pi k) + k(\log k - 1) + \frac{1}{12k}$. Taking logarithms in the definition of $\widehat{C}_n$ (all factors are positive: $U(q) > 0$, $\log(2+\sqrt{5}) > 0$),

$$\begin{aligned}\log \widehat{C}_n &= \log 2 + \frac{m}{2}\log\frac{4}{\pi} + \log k! + m\log\frac{U(q)}{\log(2+\sqrt{5})}\\ &\le \log 2 + \frac{m}{2}\log\frac{4}{\pi} + \frac{1}{2}\log(2\pi k) + k(\log k - 1) + \frac{1}{12k}\\ &\quad + m\log\frac{U(q)}{\log(2+\sqrt{5})} \;=\; T_n.\end{aligned}$$

If $d_n(\ell)\log\ell > T_n$ then $\ell^{d_n(\ell)} = e^{d_n(\ell)\log\ell} > e^{T_n} \ge \widehat{C}_n$, and Corollary 5.11 gives $\ell \nmid k_n$.

Check at $n = 7$. $m = 64$, $k = 33$, $q = 512$, $U(512) = 2.2431\ldots$: $T_7 = \log 2 + 32\log(4/\pi) + \frac{1}{2}\log(66\pi) + 33(\log 33 - 1) + \frac{1}{396} + 64\log(2.24312/1.44364) = 0.69315 + 7.73006 + 2.66719 + 82.38475 + 0.00253 + 28.20496 = 121.68264$, while $\log\widehat{C}_7 = 121.68264$ ($\widehat{C}_7 = 7.016 \times 10^{52}$, `sage/r16_hatCn/r16_hatCn_pilot.log`); $T_7 - \log\widehat{C}_7 = 7.7 \times 10^{-8}$, i.e. $1/(12k) - r_{33}$ with r_{33} the Robbins remainder ($1/397 < r_{33} < 1/396$). Every constant is displayed in T_n; the remark that $T_n = \frac{m}{2}\log m + m\log(n+2) + O(m)$ (from $\log k! = k\log k - k + O(\log k)$ and $U(2^{n+2}) = \frac{(n+2)\log 2}{4} + O(1)$) is descriptive and is not part of the statement. ☐

Lemma 5.13 (C7: rigorous enclosure of C_7 [C])**.** *$C_7 \in [C_7^-, C_7^+]$, where both endpoints are recorded to 160 significant digits, rounded outward, in `certificates/constants/Cn_interval_r14.json` (digamma route, 500-bit ball arithmetic, `sage/r14_cn_interval.sage/.log`; $C_7^+ - C_7^- < 5 \times 10^{-115}$; the two endpoints share their first 145 digits). The displayed value $C_7 = 1.7273421630363529579743237623519834054\ldots \times 10^{30}$ is the truncation of that common prefix to 38 digits (truncation error $< 10^{-7}$; the width of the enclosure is a property of the certificate, not of the displayed digits). The enclosure is consistent with the interval determinant of `certificates/blichfeldt/r6_blichfeldt_cert.log`, the PARI Gauss-sum route and the Magma evaluation (the paper's Section "Evaluation of the constant"); in particular $C_7^+ < T_1 = 1727342163036353095979941756929$ (asserted inside the certificate against the frozen integer of Lemma 5.14).*

Lemma 5.14 (C7 integer comparison [F])**.** *For integers ℓ:*

$$1314283897427172 < \ell \implies 1727342163036353095979941756929 < \ell^2.$$

Proof. The degree-1 statement is the hypothesis itself ($\ell^1 = \ell$). For degree 2: write $T_2 = 1314283897427172$. If $\ell > T_2$ then $\ell \ge T_2 + 1$, so $(T_2+1)^2 \le \ell^2$ by monotonicity of squaring on $\mathbb{N}$, and

$$1314283897427173^2 = 1727342163036359799448838771929 > 1727342163036353095979941756929$$

by exact integer multiplication; hence $1727342163036353095979941756929 < \ell^2$. The real-valued form used in Corollary 5.15 follows by casting $\mathbb{N} \to \mathbb{R}$. (Lean: `deg1_threshold_valid`, `deg2_threshold_valid`, `c7_deg2_instance`; the only computation is this one exact square, discharged by `norm_num`.) The Lean declaration(s) named above are this proof verbatim; `#print axioms` within std-3 (lean/README_lean.md). ☐

Corollary 5.15 (n=7 [C + F])**.**

$$C_7 = 1.7273421630363529579743237623519834054\ldots \times 10^{30}$$

(the 38 digits shown are a truncation of the certified enclosure of the paper's Section "Evaluation of the constant", truncation error $< 10^{-7}$; no other error is attached to the displayed digits). Hence $\ell \nmid h_7/h_6$ whenever $\ell \equiv 1 \pmod{128}$ and $\ell > 1727342163036353095979941756929$, or $\ell \equiv 63, 65, 127 \pmod{128}$ and $\ell > 1314283897427172$.

Proof. Take $n = 7$, $m = 64$, $k_7 = h_7/h_6$. By Lemma 3.3, $d_7(\ell) = 1$ for $\ell \equiv 1 \pmod{128}$ and $d_7(\ell) = 2$ for $\ell \equiv 63, 65, 127 \pmod{128}$. Lemma 5.13 encloses C_7 in a rigorous ball whose upper end is below the integer $T_1 = 17273421630363530959799417756929$, and $T_2 = 1314283897427172$ satisfies $T_1 < \ell^2$ whenever $T_2 < \ell$ (Lemma 5.14, kernel-checked; the comparison $C_7 < T_1$ and ${T_2}^2$ against the ball are verifier step 04). Hence $\ell > T_1$ gives $\ell^1 > C_7$, and $\ell > T_2$ with $d_7(\ell) = 2$ gives $\ell^2 > T_1 > C_7$; in both cases Theorem 5.2 yields $\ell \nmid k_7$. $\square$

Chapter 6

Scaling (Appendix E of the paper)

Lemma 6.1 (S-group: $L \cap kR = kL$ for k coprime to the index [F]). *R abelian, $[R : L] < \infty$, $\gcd(k, [R : L]) = 1 \Rightarrow L \cap kR = kL$.*

Proof. Let R be an abelian group (written additively), $L \leq R$ a subgroup of finite index $N = [R : L]$, and $k \geq 1$ an integer with $\gcd(k, N) = 1$. We show $L \cap kR = kL$, where $kR = \{kr : r \in R\}$ and $kL = \{ky : y \in L\}$.

$kL \subseteq L \cap kR$. If $x = ky$ with $y \in L$, then $x \in kR$ by definition, and $x \in L$ because a subgroup is closed under integer multiples.

$L \cap kR \subseteq kL$. Let $x \in L \cap kR$, say $x = kr$ with $r \in R$, and let $\pi : R \to R/L$ be the projection. Since π is a homomorphism, $\pi(x) = k\,\pi(r)$, and $\pi(x) = 0$ because $x \in L$; hence $k\,\pi(r) = 0$ in the finite group R/L of order N. Multiplication by k is injective on R/L: if $k\bar{y} = 0$, choose integers a, b with $ak + bN = 1$ (possible since $\gcd(k, N) = 1$); then $\bar{y} = (ak + bN)\bar{y} = a(k\bar{y}) + b(N\bar{y}) = 0$, because $N\bar{y} = 0$ by Lagrange's theorem ($\bar{y}$ has order dividing $|R/L| = N$). (This is `coprime_smul_injective`: on a finite abelian group whose cardinality is coprime to k, $y \mapsto ky$ is injective.) Applied to $\bar{y} = \pi(r)$ it gives $\pi(r) = 0$, i.e. $r \in L$, and therefore $x = kr \in kL$.

The two inclusions together are the equivalence $kr \in L \iff r \in L$ for every $r \in R$ (`smul_mem_iff_of_coprime`, whose hypotheses are exactly: R an additive commutative group, L a subgroup with R/L finite, and $\gcd(k, |R/L|) = 1$). In the application (Theorem 6.5) $R = R_n$, $L = L_f$ (notation of the setting), $[R_n : L_f]$ is a power of the odd prime ℓ and $k = 2^{t-1}$, so $\gcd(k, [R_n : L_f]) = 1$ and $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$. $\square$

Lemma 6.2 (S-analytic: the general-t inequality [F]). *For every $t \geq 2$: $(2 + \sqrt{5})^{2^{t-1}} > 2^t + \sqrt{4^t + 1}$.*

Proof. Four steps, each an elementary real inequality. (1) $\sqrt{4^t + 1} < 2^t + 1$: both sides are positive and $(2^t + 1)^2 = 4^t + 2^{t+1} + 1 > 4^t + 1$. Hence $2^t + \sqrt{4^t + 1} < 2^{t+1} + 1$. (2) For $t \geq 2$, $2^{t+1} + 1 \leq 4^t$: write $4^t = (2^t)^2$ and $2^{t+1} = 2 \cdot 2^t$; with $x = 2^t \geq 4$ the claim is $2x + 1 \leq x^2$, i.e. $(x - 1)^2 \geq 2$, true for $x \geq 4$. (3) $4^t \leq 4^{2^{t-1}}$ because $t \leq 2^{t-1}$ for $t \geq 1$ (induction: $1 \leq 1$, and $s + 1 < 2^s$ for $s \geq 1$ gives $s + 2 \leq 2^{s+1}$). (4) $4^{2^{t-1}} < (2 + \sqrt{5})^{2^{t-1}}$ because $4 < 2 + \sqrt{5}$ (as $2 = \sqrt{4} < \sqrt{5}$) and the exponent 2^{t-1} is positive. Chaining (1)–(4): $2^t + \sqrt{4^t + 1} < 2^{t+1} + 1 \leq 4^t \leq 4^{2^{t-1}} < (2 + \sqrt{5})^{2^{t-1}}$. (Lean: `ineq_t_general` with the helpers `sqrt_four_pow_add_one_lt` and `le_two_pow_pred`; the instances $t = 2, 3, 4$ are `ineq_t2'/t3'/t4'` and agree with the independent earlier proofs `ineq_t2/t3/t4`.) The Lean declaration(s) named above are this proof verbatim; `#print axioms` within std-3 (lean/README_lean.md). $\square$

Lemma 6.3 (depth-t height floor [M, L-relative]). *For $t \geq 1$ and $u \in RE_n^+ \setminus \{\pm 1\}$ with $u \equiv 1 \pmod{2^t \mathcal{O}_n}$: $\mathrm{ht}(u) \geq L_{n,t} = \sqrt{2^n}\,\mathrm{arcsinh}(2^t)$.*

Note. $L_{n,1} = L_n$. This is the floor used by the depth-t variant in Theorem 6.5; from r16 it is a lemma, not a hypothesis of that theorem.

Proof. Let $t \geq 1$ and $u \in RE_n^+ \setminus \{\pm 1\}$ with $u \equiv 1 \pmod{2^t \mathcal{O}_n}$. By Lemma 3.1, $u \notin B_{n-1}$; since $B_n/\mathbb{Q}$ is cyclic of 2-power degree, its subfields are the B_k, so $\mathbb{Q}(u) = B_n$ and u has degree $N = 2^n$. From $u-1 \in 2^t\mathcal{O}_n$ and $u+1 = (u-1)+2 \in 2\mathcal{O}_n$ we get $u^2-1 \in 2^{t+1}\mathcal{O}_n$, so the ideal $\mathfrak{M} = 2^{t+1}\mathcal{O}_n$ contains u^2-1 and has absolute norm $C = 2^{(t+1)N}$, $C^{1/N} = 2^{t+1}$. Schinzel's inequality in the form quoted as [15] Thm. 5.1 ("Let ε be a totally real unit different from ± 1 and $\mathfrak{M}$ an ideal of $\mathbb{Q}(\varepsilon)$ containing $\varepsilon^2 - 1$; then $M(\varepsilon) \geq \big((C^{1/d} + \sqrt{C^{2/d}+4})/2\big)^{d/2}$, C the absolute norm of $\mathfrak{M}$, d the degree") gives, with $d = N$, $M(u) \geq \big((2^{t+1} + \sqrt{4^{t+1}+4})/2\big)^{N/2} = \big(2^t + \sqrt{4^t+1}\big)^{N/2} = \exp\big(\frac{N}{2}\operatorname{arcsinh}(2^t)\big)$, using $\operatorname{arcsinh} x = \log(x + \sqrt{x^2+1})$. Finally, for a unit the conjugates satisfy $\sum_i \log|u_i| = 0$, so $\log M(u) = \sum_{|u_i|>1} \log|u_i| = \frac{1}{2}\sum_i |\log|u_i|| \leq \frac{1}{2}\sqrt{N}\,\mathrm{ht}(u)$ by Cauchy–Schwarz (this is [14] eq. (2.4), proof of their Lemma 2.3). Hence $\mathrm{ht}(u) \geq \frac{2}{\sqrt{N}} \cdot \frac{N}{2}\operatorname{arcsinh}(2^t) = \sqrt{2^n}\operatorname{arcsinh}(2^t) = L_{n,t}$. For $t = 1$ this is $\sqrt{2^n}\log(2+\sqrt{5}) = L_n$, because $\operatorname{arcsinh} 2 = \log(2+\sqrt{5})$. (This floor is proved here for every $t \geq 1$, so Theorem 6.5 keeps only hypothesis (i), which remains open for $t \geq 5$.) □

Lemma 6.4 (odd-power depth transfer [M, F-core])**.** *Let $t \geq 1$, ℓ an odd integer and $x, y \in \mathcal{O}_n^\times$. Then $x^\ell \equiv 1 \pmod{2^t\mathcal{O}_n}$ if and only if $x \equiv 1 \pmod{2^t\mathcal{O}_n}$; more generally $x^\ell \equiv y^\ell \pmod{2^t\mathcal{O}_n}$ if and only if $x \equiv y \pmod{2^t\mathcal{O}_n}$.*

Note. Inputs proved inside the lemma: 2 is totally ramified in B_n with residue field $\mathbb{F}_2$, so $(\mathcal{O}_n/2^t\mathcal{O}_n)^\times$ is a group of order $2^{2^n t-1}$; the odd power map on a group of 2-power order is a bijection (`lean/WeberOddTransfer.lean`, five declarations, std-3; r15, GPT r14 sect 4, E15-1). The manuscript's identification of $(\mathcal{O}_n/2^t\mathcal{O}_n)^\times$ as a 2-group is [M]; only the group statement is kernel-checked.

Proof. Write $N = 2^n = [B_n : \mathbb{Q}]$, $q = 2^{n+2}$, $\zeta = e^{2\pi i/q}$, $K = \mathbb{Q}(\zeta)$ and $\pi = 1 - \zeta$; $B_n = \mathbb{Q}(\zeta + \zeta^{-1}) \subset K$ with $[K : B_n] = 2$.

Step 1: 2 is totally ramified in B_n with residue field $\mathbb{F}_2$. For odd a with $0 < a < q$ the quotient $(1-\zeta^a)/(1-\zeta) = 1 + \zeta + \cdots + \zeta^{a-1}$ lies in $\mathbb{Z}[\zeta]$, and if $ab \equiv 1 \pmod q$ then $(1-\zeta)/(1-\zeta^a) = (1-\zeta^{ab})/(1-\zeta^a) = 1 + \zeta^a + \cdots + \zeta^{a(b-1)} \in \mathbb{Z}[\zeta]$; so $1 - \zeta^a = \pi\eta_a$ with $\eta_a \in \mathbb{Z}[\zeta]^\times$. The q-th cyclotomic polynomial is $\Phi_q(X) = X^{q/2}+1 = \prod_{a \text{ odd}}(X - \zeta^a)$, hence $\prod_{a \text{ odd}}(1-\zeta^a) = \Phi_q(1) = 2$, i.e. $2 = \eta\,\pi^{\varphi(q)}$ with η a unit and $\varphi(q) = 2^{n+1} = [K : \mathbb{Q}]$. Since the ramification index of a prime above 2 is at most $[K : \mathbb{Q}]$, this forces $2\mathcal{O}_K = \mathfrak{P}^{[K:\mathbb{Q}]}$ with $\mathfrak{P} = \pi\mathcal{O}_K$ the unique prime of K above 2, and its residue degree is 1. Let $\mathfrak{p} = \mathfrak{P} \cap \mathcal{O}_n$. Every prime of $\mathcal{O}_n$ above 2 lies below $\mathfrak{P}$, so $\mathfrak{p}$ is the unique such prime, and ramification indices multiply in the tower $\mathbb{Q} \subset B_n \subset K$: $[K : \mathbb{Q}] = e(\mathfrak{P}/2) = e(\mathfrak{P}/\mathfrak{p})\,e(\mathfrak{p}/2) \leq [K : B_n]\,[B_n : \mathbb{Q}] = [K : \mathbb{Q}]$, whence $e(\mathfrak{p}/2) = [B_n : \mathbb{Q}] = N$. Then $2\mathcal{O}_n = \mathfrak{p}^N$, the residue degree $f(\mathfrak{p}/2)$ is 1 (because $ef \leq N$), $\mathcal{O}_n/\mathfrak{p} = \mathbb{F}_2$, and $2^t\mathcal{O}_n = \mathfrak{p}^{Nt}$ for every $t \geq 1$.

Step 2: the unit group of $\mathcal{O}_n/2^t\mathcal{O}_n$ is a 2-group. Put $k = Nt \geq 1$. The ring $\bar{R} = \mathcal{O}_n/\mathfrak{p}^k$ is finite of order $\mathrm{N}(\mathfrak{p}^k) = \mathrm{N}(\mathfrak{p})^k = 2^k$ (multiplicativity of the absolute norm in the Dedekind domain $\mathcal{O}_n$), it is local with unique maximal ideal $\bar{\mathfrak{p}} = \mathfrak{p}/\mathfrak{p}^k$ (every ideal of $\bar{R}$ is $\mathfrak{p}^i/\mathfrak{p}^k$, $0 \leq i \leq k$), and $|\bar{\mathfrak{p}}| = \mathrm{N}(\mathfrak{p})^{k-1} = 2^{k-1}$. In a finite local ring the units are exactly the elements outside the maximal ideal, so $|\bar{R}^\times| = 2^k - 2^{k-1} = 2^{k-1}$.

Step 3: the odd power map on $\bar{R}^\times$ is a bijection. **[F]**: `WeberOddTransfer.pow_odd_eq_one_iff_of_two_pow_card` (a group G with $|G| = 2^{k'}$ and ℓ odd satisfy $g^\ell = 1 \iff g = 1$), `pow_odd_eq_iff_of_two_pow_card` ($g^\ell = h^\ell \iff g = h$) and the ring form `isUnit_pow_odd_eq_one_iff`; the proof is Mathlib's `powCoprime`: when $\gcd(|G|, \ell) = 1$ the map $g \mapsto g^\ell$ is a bijection of G with inverse $g \mapsto g^b$, $b\ell \equiv 1 \pmod{|G|}$. Here $|\bar{R}^\times| = 2^{k-1}$ and ℓ is odd.

Step 4: transfer. Let $x \in \mathcal{O}_n^\times$ and let $\bar{x}$ be its image in $\bar{R} = \mathcal{O}_n/2^t\mathcal{O}_n$; $\bar{x} \in \bar{R}^\times$ because x is a unit, and for any $y \in \mathcal{O}_n$ one has $y \equiv 1 \pmod{2^t\mathcal{O}_n}$ iff $\bar{y} = 1$. Hence $x^\ell \equiv 1 \pmod{2^t\mathcal{O}_n} \iff \bar{x}^\ell = 1 \iff \bar{x} = 1 \iff x \equiv 1 \pmod{2^t\mathcal{O}_n}$, the middle equivalence being Step 3. For units x, y the same argument applied to $xy^{-1} \in \mathcal{O}_n^\times$ gives $x^\ell \equiv y^\ell \iff x \equiv y \pmod{2^t\mathcal{O}_n}$.

Remark (valuation form of the same fact). Since $\mathcal{O}_n/\mathfrak{p} = \mathbb{F}_2$, every unit satisfies $x \equiv 1 \pmod{\mathfrak{p}}$, so $1 + x + \cdots + x^{\ell-1} \equiv \ell \equiv 1 \pmod{\mathfrak{p}}$ is a $\mathfrak{p}$-adic unit and $v_{\mathfrak{p}}(x^\ell - 1) = v_{\mathfrak{p}}(x-1) +$

$v_{\mathfrak{p}}(1+x+\cdots+x^{\ell-1}) = v_{\mathfrak{p}}(x-1)$; the two congruences of the lemma are the two conditions $v_{\mathfrak{p}}(\cdot) \geq Nt$ on these equal quantities. This second proof is recorded for the reader; the proof above is the one whose discrete core is kernel-checked. $\square$

Theorem 6.5 (S0: abstract scale law [M relative to (i); F-core]). *Let $t \geq 2$ and let $J_t = \{a \in R_n : u_a \equiv 1 \pmod{2^t}\}$ where $u_a = \prod_i \sigma^i(\varepsilon_n)^{a_i}$, and put $L_{n,t} := \sqrt{2^n}\,\mathrm{arcsinh}(2^t)$. Assume (i) $J_t = 2^{t-1}R_n$. Then $\Lambda_{f,t} := \ell^{-1}H_n(L_f \cap J_t) = 2^{t-1}\Lambda_f$ for every component f, and the depth-t variant of Theorem A (Blichfeldt on $\Lambda_{f,t}$ against the depth-t floor $L_{n,t}$ of Lemma 6.3, applied to the unit that carries the short vector) has a stricter sufficient condition than Theorem A, i.e. the resulting direct filtered-lattice criterion is strictly weaker, because $(2+\sqrt{5})^{2^{t-1}} > 2^t + \sqrt{4^t+1}$ for every $t \geq 2$.*

Note. r15: the former hypothesis (ii) is now Lemma 6.3; the transfer from the depth of $u_a = r_a^\ell$ to the depth of r_a is Lemma 6.4 (E15-1); the floor is applied to r_a in the non-base branch and to u_b in the base branch.

Proof. The lattice half. R_n/L_f is an $\mathbb{F}_\ell$-vector space (Lemma 3.4), so $[R_n : L_f]$ is a power of the odd prime ℓ and Lemma S1 (the group lemma) applies with $k = 2^{t-1}$: $L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$; likewise $\ell R_n \cap 2^{t-1}R_n = 2^{t-1}\ell R_n$. Under (i), $L_f \cap J_t = L_f \cap 2^{t-1}R_n = 2^{t-1}L_f$, hence $\Lambda_{f,t} = \ell^{-1}H_n(2^{t-1}L_f) = 2^{t-1}\Lambda_f$, and the vectors of $\Lambda_{f,t}$ coming from $a \notin \ell R_n$ are exactly the 2^{t-1}-dilates of those of Λ_f coming from $a \notin \ell R_n$ (the same lemma for ℓR_n).

What the depth-t variant needs. The depth-t variant of Theorem A is the argument of Prop. F with Λ_{f_0} replaced by $\Lambda_{f_0,t}$ and the floor L_n of Lemma A$^+$ replaced by the depth-t floor $L_{n,t}$ of Lemma 6.3. We make explicit the unit to which the floor is applied, in both branches of Prop. F. Suppose $\ell \mid k_n$, let f_0 be the saturation component of Lemma 3.6, and let $v \in \Lambda_{f_0,t}$ be a nonzero vector; write $v = \ell^{-1}H_n a$ with $a \in L_{f_0} \cap J_t$, $a \neq 0$. Since $a \in J_t$, $u_a \equiv 1 \pmod{2^t\mathcal{O}_n}$. (Non-base branch, $a \notin \ell R_n$.) By Lemma 3.6, $r_a \in RE_n^+$ and v is the log vector of r_a, so $\mathrm{ht}(r_a) = |v|$; $r_a \neq \pm 1$ because $v \neq 0$. The unit r_a is a formal ℓ-th root: $r_a^\ell = u_a \equiv 1 \pmod{2^t\mathcal{O}_n}$. Lemma 6.4 (odd-power depth transfer, $x = r_a$) gives $r_a \equiv 1 \pmod{2^t\mathcal{O}_n}$, and Lemma 6.3 gives $\mathrm{ht}(r_a) \geq L_{n,t}$. (Base branch, $a = \ell b$, $b \in R_n \setminus \{0\}$.) By Lemma 3.7, $v = H_n b$ is the log vector of $u_b \in A_n \subseteq RE_n^+$ and $u_b \neq \pm 1$; here $u_b^\ell = u_{\ell b} = u_a \equiv 1 \pmod{2^t\mathcal{O}_n}$, so Lemma 6.4 ($x = u_b$) gives $u_b \equiv 1 \pmod{2^t\mathcal{O}_n}$ and Lemma 6.3 gives $\mathrm{ht}(u_b) \geq L_{n,t}$. In both branches the unit carrying v has height $|v| \geq L_{n,t}$. Therefore the depth-t variant reaches its contradiction exactly when Blichfeldt supplies a nonzero $v \in \Lambda_{f_0,t}$ with $|v| < L_{n,t}$; without Lemma 6.4 the passage from the depth of $u_a = r_a^\ell$ (the quantity filtered by J_t) to the depth of r_a (the unit whose height is bounded) would be unjustified.

Comparison. By the lattice half, a nonzero $v \in \Lambda_{f_0,t}$ with $|v| < L_{n,t}$ is $v = 2^{t-1}v'$ with a nonzero $v' \in \Lambda_{f_0}$ and $|v'| < L_{n,t}/2^{t-1}$, whereas Theorem A (Prop. F) needs a nonzero $v' \in \Lambda_{f_0}$ with $|v'| < L_n$. Now $L_{n,t}/2^{t-1} < L_n$ iff $\mathrm{arcsinh}(2^t) < 2^{t-1}\log(2+\sqrt{5})$ iff $2^t + \sqrt{4^t+1} < (2+\sqrt{5})^{2^{t-1}}$, and the last inequality holds for every $t \geq 2$ (Lemma S-analytic, **[F]**: `ineq_t_general`: $2^t + \sqrt{4^t+1} < 2^{t+1}+1 \leq 4^t \leq 4^{2^{t-1}} < (2+\sqrt{5})^{2^{t-1}}$). Hence every vector that satisfies the depth-t requirement satisfies the requirement of Theorem A, and the requirement of the depth-t variant is strictly stronger: the direct filtered-lattice sufficient condition at depth t is stricter than that of Theorem A, i.e. the resulting direct filtered-lattice criterion is weaker. This is the only sense of "weaker" used in this paper; it concerns the direct application of the same Blichfeldt argument to the deeper congruence sublattice $\Lambda_{f,t}$ and says nothing about other 2-adic refinements. Nothing is asserted about the truth of hypothesis (i) beyond the certified cases of Cor. S1. $\square$

Lemma 6.6 (J_t certificates, $t = 2,3,4$ [C]). *$J_t = 2^{t-1}R_n$ at $n = 7$ for $t = 2,3,4$ (`certificates/twoadic_rank/fam{1,2,3}_*.txt`; `sage/r9_scaling_no_go.log`): constructed independently in Sage and in Magma, and replayed by the standalone Python rank* *checker* ***scripts/verify_ twoadic_ rank. py*** *(a replay of the stored bit matrices, not a third construction; E15-2).*

Corollary 6.7 (S1 [C + F])**.** *Hypothesis (i) holds for* $t = 2, 3, 4$ *at* $n = 7$ *(exact rank certificates). Hence the direct filtered-lattice routes at depths* 4*,* 8 *and* 16 *(Blichfeldt on* $\Lambda_{f,t}$ *against* $L_{n,t}$*) are strictly weaker than Theorem A. No statement is made about other* 2*-adic refinements.*

Proof. For $t = 2, 3, 4$ at $n = 7$ the equality $J_t = 2^{t-1}R_n$ was established by exact rank computations of the 2-adic filtration of A_n, constructed independently in Sage (`sage/r8_twoadic_depth_exact.sage`) and in Magma (`sage/r8_twoadic_rank_magma.m`), and replayed by a standalone Python rank checker that recomputes the rank of the saved bit matrices (`scripts/verify_twoadic_rank.py`, certificates in `certificates/twoadic_rank/`; log `sage/r9_scaling_no_go.log`); the Python step is a replay of stored certificates, not a third construction. Lemma 6.4 and Lemma 6.3 hold for every $t \geq 1$ (the latter relative to its literature inputs, Schinzel and MO16 eq. (2.4)), so Theorem 6.5 applies at $t = 2, 3, 4$ and the direct filtered-lattice routes at depths 4, 8 and 16 are strictly weaker than Theorem A. □

Chapter 7

The cyclotomic $\mathbb{Z}_3$-tower (Track B)

Definition 7.1 (The $p = 3$ data)**.** $B_{3,n}$ the degree-3^n layer, $N = 3^n$, $q = 3^{n+1}$, $\eta_n = \sin(2(1 + 3^n)\pi/q)/\sin(2\pi/q)$ the Horie unit (MO13 §1, one factor at $p = 3$), σ the generator of $\mathrm{Gal}(B_{3,n}/\mathbb{Q})$ induced by $\zeta_q \mapsto \zeta_q^4$, $\tau = \sigma^{3^{n-1}}$; for $\ell \neq 3$: $f = \mathrm{ord}_3(\ell)$, $3^s \| \ell^2 - 1$, $r = \min(n, s)$, $c = 2 \cdot 3^{r-1}$; $u_\alpha = \eta_n^{\alpha_\sigma}$ for $\alpha \in \mathbb{Z}[\zeta_{3^r}] \subseteq \mathbb{Z}[\zeta_{3^n}]$ (lift exponents $3^{n-r}j$, $j < c$); $\lambda_j = \log|\sigma^j \eta_n|$, $\hat{\lambda}_k = \sum_j \lambda_j e^{-2\pi ijk/N}$; $D_r^{(n)}$ the covolume of the lattice spanned by $H(\sigma^{3^{n-r}j}\eta_n)$, $j < c$; $U_{3,n} = \{\varepsilon \in E_{3,n} : \mathrm{Nr}_{B_{3,n}/B_{3,n-1}}\, \varepsilon = 1\}$; $L_{3,n}^{\mathrm{rel}} = \sqrt{N} \log \frac{3^{(N-1)/(2N)} + \sqrt{3^{(N-1)/N} + 4}}{2}$; $C_{n,r}^{(3)} = (2/\pi)^{c/2}\Gamma(2 + c/2)D_r^{(n)}/(L_{3,n}^{\mathrm{rel}})^c$. Throughout the $\mathbb{Z}_3$-tower, "$\ell \mid h_{3,n}/h_{3,n-1}$" means $v_\ell(h_{3,n}) > v_\ell(h_{3,n-1})$, and "$\ell \nmid h_{3,n}/h_{3,n-1}$" its negation; by Lemma 4.1 with $p = 3$ and $\ell \neq 3$ (the extension $B_{3,n}/B_{3,n-1}$ is cyclic of degree 3) this is $\ker(N)_\ell \neq 0$, and $v_\ell(h_{3,n}) \geq v_\ell(h_{3,n-1})$ always holds, so the quotient is ℓ-integral for every $\ell \neq 3$.

Lemma 7.2 (Relative-norm-one height floor [L])**.** *If $\varepsilon \in E_{3,n} \setminus E_{3,n-1}$ and $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$, then $\mathrm{ht}(\varepsilon) \geq L_{3,n}^{\mathrm{rel}} := \sqrt{3^n} \log \frac{3^{(3^n-1)/(2\cdot 3^n)} + \sqrt{3^{(3^n-1)/3^n} + 4}}{2}$, where ht is the Euclidean norm of the log vector ([14] Def. 2.2). ([14] Lemma 2.5(2): Lemma 2.3 there, the Mahler-measure bound with $C = |\mathrm{Nr}_{B_{3,n}/\mathbb{Q}}(\varepsilon^2 - 1)|$, together with Lemma 2.4(2), $C \geq 3^{(3^n-1)/2}$, which is [10] Lemma 9.1.)*

Note. Read verbatim from the JTNB PDF (`paper/MO2016_height_weber_jtnb965.pdf`).

Lemma 7.3 (Relative norm of the Horie unit [F])**.** *$a = 1 + 3^n$ has exact order 3 modulo q and τ acts by $\zeta \mapsto \zeta^a$; $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\eta_n) = 1$. If $\eta_n^{\alpha_\sigma} = \varepsilon^\ell$ with $\alpha_\sigma \in \mathbb{Z}[G]$, $\varepsilon \in E_{3,n}$, ℓ an odd prime, then $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$. In general, for $\varepsilon \in E_{3,n-1}$ one has $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = \varepsilon^3$, so a unit of the lower layer with relative norm 1 equals 1; hence a unit of relative norm 1 with $H(\varepsilon) \neq 0$ is not in $E_{3,n-1}$.*

Note. Every such ε lies in $U_{3,n} \setminus E_{3,n-1}$ and satisfies the hypothesis of Lemma 7.2.

Proof. Let $N = 3^n$, $q = 3N$, $\zeta = \zeta_q$, $a = 1 + N$ and $\tau = \sigma^{3^{n-1}}$, where σ is induced by $\zeta \mapsto \zeta^4$.

(i) a has exact order 3 modulo q. $(1 + N)^3 - 1 = 3N + 3N^2 + N^3 = q\,(1 + N + N^2/3)$ with $3 \mid N^2$ for $n \geq 1$, so $a^3 \equiv 1 \pmod{q}$ (`WeberP3Rel.a_cube_modEq_one`); and $a \not\equiv 1 \pmod{q}$ since $0 < N < q$, so the order is exactly 3 (the only divisors of 3 being 1 and 3).

(ii) τ acts by $\zeta \mapsto \zeta^a$. By induction $4^{3^k} = 1 + 3^{k+1} + 3^{k+2}t_k$ with $t_k \in \mathbb{Z}$ (from $(1 + u)^3 = 1 + 3u + 3u^2 + u^3$, $u = 3^{k+1}(1 + 3t_{k-1})$), hence $4^{3^{n-1}} \equiv 1 + 3^n = a \pmod{3^{n+1}}$ (`WeberP3Rel.four_pow_modEq`). Since $G = \mathrm{Gal}(B_{3,n}/\mathbb{Q})$ is cyclic of order N and τ has order 3 by (i), τ generates its unique subgroup of order 3, which is $\mathrm{Gal}(B_{3,n}/B_{3,n-1})$.

(iii) Relative norm. For t prime to 3, $\sigma_t : \zeta \mapsto \zeta^t$ sends $\eta_n = (\zeta^a - \zeta^{-a})/(\zeta - \zeta^{-1}) = \sin(2a\pi/q)/\sin(2\pi/q)$ to $\sin(2ta\pi/q)/\sin(2t\pi/q)$. With $s(i) = \sin(2a^i\pi/q)$ ($i = 0, 1, 2, 3$; all nonzero since $q \nmid 2a^i$) this gives $\tau^i\eta_n = s(i + 1)/s(i)$ and

$$\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\eta_n) = \eta_n \cdot \tau\eta_n \cdot \tau^2\eta_n = \frac{s(1)}{s(0)}\frac{s(2)}{s(1)}\frac{s(3)}{s(2)} = \frac{s(3)}{s(0)} = 1,$$

because $a^3 = 1 + mq$ gives $s(3) = \sin(2\pi/q + 2\pi m) = s(0)$ (`WeberP3Rel.sin_telescope_step`, `WeberP3Rel.telescope_three`).

(iv) Transfer to the saturation root. Let $\alpha_\sigma \in \mathbb{Z}[G]$ and $\varepsilon \in E_{3,n}$ with $\eta_n^{\alpha_\sigma} = \varepsilon^\ell$, ℓ an odd prime. As G is abelian, Nr commutes with α_σ, so $\mathrm{Nr}(\varepsilon)^\ell = \mathrm{Nr}(\eta_n)^{\alpha_\sigma} = 1$; $\mathrm{Nr}(\varepsilon)$ is a real number ($B_{3,n-1}$ is totally real) and ℓ is odd, hence $\mathrm{Nr}(\varepsilon) = 1$ (`WeberP3Rel.eq_one_of_odd_pow_eq_one`). If moreover $H(\varepsilon) \neq 0$ then $\varepsilon \notin E_{3,n-1}$ by (v) below. Thus every such ε with $H(\varepsilon) \neq 0$ lies in $E_{3,n} \setminus E_{3,n-1}$ and has relative norm 1; the subgroup $U_{3,n} := \{\varepsilon \in E_{3,n} : \mathrm{Nr}_{B_{3,n}/B_{3,n-1}} \varepsilon = 1\}$ contains all ℓ-th roots produced by Horie's lemma ([10] Lemma 1.3).

(v) Units of the lower layer with relative norm 1. Let $\varepsilon \in E_{3,n-1}$. The Galois group of $B_{3,n}/B_{3,n-1}$ is $\langle\tau\rangle$, of order 3, and it fixes ε, so

$$\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = \varepsilon \cdot \tau\varepsilon \cdot \tau^2\varepsilon = \varepsilon^3.$$

Hence $\mathrm{Nr}(\varepsilon) = 1$ gives $\varepsilon^3 = 1$ with ε real, so $\varepsilon = 1$ (`WeberP3Rel.eq_one_of_odd_pow_eq_one` with the odd exponent 3). Consequently a unit $\varepsilon \in U_{3,n}$ with $H(\varepsilon) \neq 0$ cannot lie in $E_{3,n-1}$: it would equal 1 and have $H(\varepsilon) = 0$. The Lean declarations named above are the five discrete steps (order of a, action of τ, telescoping, periodicity of sin, odd-power transfer) verbatim; the Galois-theoretic sentences ((ii) second half, (iii) first sentence, (iv) commutation of Nr with $\mathbb{Z}[G]$) are M. $\square$

Lemma 7.4 (Discriminant of $\mathbb{Q}(\zeta_{p^r})$ [L])**.** *For a prime p and $r \geq 1$, $|\operatorname{disc}\mathbb{Q}(\zeta_{p^r})| = p^{p^{r-1}(pr-r-1)}$ and $\mathbb{Z}[\zeta_{p^r}]$ is the ring of integers of $\mathbb{Q}(\zeta_{p^r})$ ([20], Chapter 2). In particular $|\operatorname{disc}\mathbb{Q}(\zeta_{3^r})| = 3^{3^{r-1}(2r-1)}$, and this equals $\prod_{b\neq b'}|\xi^b - \xi^{b'}|$ over the $\varphi(3^r)$ primitive 3^r-th roots of unity ξ^b, $b \in (\mathbb{Z}/3^r)^\times$, because the discriminant of the power basis $1, \zeta, \ldots, \zeta^{\varphi(3^r)-1}$ is $\operatorname{disc}\Phi_{3^r} = \prod_{b\neq b'}(\xi^b - \xi^{b'})$ and the power basis is an integral basis.*

Note. Washington is cited at chapter level; no theorem number is load-bearing (hw 29/984).

Theorem 7.5 (Spectral rank and exact covolume [F-core; L for $L(1,\chi) \neq 0$])**.** *Let $\lambda_j = \log|\sigma^j\eta_n|$ ($j \in \mathbb{Z}/N$) and $\hat\lambda_k = \sum_j \lambda_j e^{-2\pi i jk/N}$. Then $\hat\lambda_k = 0$ if $3 \mid k$ and $\hat\lambda_k \neq 0$ if $3 \nmid k$. Consequently, for every $n \geq 1$ and $1 \leq r \leq n$, with $c = 2 \cdot 3^{r-1}$, the vectors $H(\sigma^{3^{n-r}i}\eta_n)$, $0 \leq i < c$, are linearly independent, and their covolume $D_r^{(n)}$ satisfies*

$$\big(D_r^{(n)}\big)^2 = 3^{3^{r-1}(2r-1)} \prod_{b\in(\mathbb{Z}/3^r)^\times} W_{n,r}(b), \qquad W_{n,r}(b) = \frac{1}{N}\sum_{k\equiv b\ (3^r)} |\hat\lambda_k|^2 > 0.$$

Note. The linear independence is (Rank) of Theorem 3.8.

Proof. Fix $n \geq 1$, $N = 3^n$, $\omega = e^{2\pi i/N}$; for $j \in \mathbb{Z}/N$ let $\lambda_j = \log|\sigma^j\eta_n|$ and $\hat\lambda_k = \sum_{j\in\mathbb{Z}/N}\lambda_j\,\omega^{-jk}$ ($k \in \mathbb{Z}/N$).

Step 1: $\hat\lambda_k = 0$ for $3 \mid k$. By Lemma 7.3(iii), $\lambda_j + \lambda_{j+N/3} + \lambda_{j+2N/3} = \log|\sigma^j \mathrm{Nr}_{B_{3,n}/B_{3,n-1}}\eta_n| = 0$ for every j ($\tau = \sigma^{N/3}$). For $k = 3k'$ one has $\omega^{-(j+N/3)k} = \omega^{-jk}$, so grouping $\mathbb{Z}/N$ into the $N/3$ cosets of $\{0, N/3, 2N/3\}$ gives $\hat\lambda_k = \sum_{j<N/3}\omega^{-jk}(\lambda_j + \lambda_{j+N/3} + \lambda_{j+2N/3}) = 0$ (`WeberP3Rel.dft_vanish_of_relnorm`).

Step 2: $\hat\lambda_k \neq 0$ for $3 \nmid k$. Since $|\sin(2\pi m/q)| = \frac{1}{2}|1 - \zeta_q^{2m}|$, $\lambda_j = g(2a \cdot 4^j) - g(2 \cdot 4^j)$ with $g(m) = \log|1 - \zeta_q^m|$ on $(\mathbb{Z}/q)^\times = \{\pm 1\} \times \langle 4\rangle$ ($q = 3N$, $a = 1 + N$, and $a \equiv 4^{N/3} \pmod q$ by Lemma 7.3(ii), `WeberP3Rel.four_pow_modEq`; the arguments of g, and below of χ_k, are residue classes modulo q). As $g(-m) = g(m)$ and $-2 \in \langle 4\rangle$, write $-2 \equiv 4^{j_0} \pmod q$; then $\lambda_j = \Gamma(j + j_0 + N/3) - \Gamma(j + j_0)$ with $\Gamma(j) = g(4^j)$, hence $\hat\lambda_k = \omega^{j_0 k}(\omega^{Nk/3} - 1)\hat\Gamma_k$ (shift rule of the twisted sum on $\mathbb{Z}/N$: `WeberP3Rel.twisted_sum_shift`, `WeberP3Rel.twisted_sum_shift_diff`) and $\omega^{Nk/3} = e^{2\pi i k/3} \neq 1$ (`WeberP3Rel.pow_third_eq_one_iff`); so $\hat\lambda_k \neq 0$ as soon as $\hat\Gamma_k \neq 0$ (`WeberP3Rel.twisted_sum_shift_diff_ne_zero`). Let χ_k be the character of $(\mathbb{Z}/q)^\times = \{\pm1\} \times \langle 4\rangle$ trivial on -1 with $\chi_k(4^j) = \omega^{-jk}$ (well defined since 4 has order N in $(\mathbb{Z}/q)^\times$ and $\omega^N = 1$); it is even. *Conductor.* Every proper divisor of $q = 3^{n+1}$ divides 3^n, so χ_k is imprimitive iff it factors through the reduction $(\mathbb{Z}/q)^\times \to (\mathbb{Z}/3^n)^\times$, i.e. iff it is trivial on the kernel of that reduction. The kernel is $\{t \in (\mathbb{Z}/q)^\times : t \equiv 1\ (3^n)\} = \{1,\, 1+3^n,\, 1+2\cdot 3^n\} = \{1, a, a^2\}$ (order 3; $a = 1+N$ has exact order 3 by Lemma 7.3(i), and $a^2 \equiv 1+2N$ as $N^2 \equiv 0 \pmod q$). Since $a \equiv 4^{N/3} \pmod q$ (Lemma 7.3(ii), `WeberP3Rel.four_pow_modEq`) and χ_k is a function of the class modulo q, $\chi_k(a) = \omega^{-Nk/3} = e^{-2\pi i k/3} \neq 1$ for $3 \nmid k$. So χ_k is not trivial on the kernel: it is primitive of conductor exactly $q = 3^{n+1}$. *Conventions.* Throughout, $\omega = e^{2\pi i/N}$ and $\hat\lambda_k = \sum_j \lambda_j \omega^{-jk}$ (minus sign in the exponent; the same sign is used in the certificate script and in the read-only verifier, and the Lean statements of `WeberP3Rel` are proved for an arbitrary ω with $\omega^N = 1$, so they cover either sign); only $|\hat\Gamma_k|$ enters below, through the absolute-value form of Washington's formula (Lemma 5.6), so no Gauss-sum normalisation is used. Then $\hat\Gamma_k = \sum_{m\in\langle 4\rangle} \chi_k(m) g(m) = \frac{1}{2}\sum_{m\in(\mathbb{Z}/q)^\times} \chi_k(m) \log|1-\zeta_q^m|$, whose absolute value is $\frac{\sqrt{q}}{2}|L(1,\chi_k)|$ by Lemma 5.6 (Washington's formula, [20], Chapter 4), and $L(1,\chi_k) \neq 0$ by Dirichlet's theorem. So $\hat\lambda_k \neq 0$.

Step 3: rank c for every n and $1 \le r \le n$. Put $h = 3^{n-r}$, $c = 2\cdot 3^{r-1} = \varphi(3^r)$ and $\xi = \omega^h$, a primitive 3^r-th root of unity. For $a_0, \dots, a_{c-1} \in \mathbb{R}$ the vector $v = \sum_{i<c} a_i H(\sigma^{hi}\eta_n)$ has coordinates $v_j = \sum_i a_i \lambda_{j+hi}$ and transform $\hat v_k = A(\omega^{hk})\hat\lambda_k$, $A(x) = \sum_i a_i x^i$. If $v = 0$ then $A(\xi^k) = 0$ for all $3 \nmid k$ by Step 2; as k runs over $(\mathbb{Z}/3^r)^\times$ the ξ^k are the c primitive 3^r-th roots of unity, and $\deg A < c$ forces $A = 0$ (`WeberP3Rel.eq_zero_of_vanish_on_primitiveRoots`, `WeberP3Rel.totient_three_pow`). Hence the c vectors $H(\sigma^{hi}\eta_n)$ are linearly independent: (Rank) holds and $D_r^{(n)} > 0$.

Step 4: exact covolume. With Parseval's identity $\langle u, v\rangle = N^{-1}\sum_k \hat u_k \overline{\hat v_k}$ the Gram matrix of the c vectors is

$$G_{ii'} = \frac{1}{N}\sum_k |\hat\lambda_k|^2\, \omega^{hk(i-i')} = \sum_{b\in(\mathbb{Z}/3^r)^\times} W_{n,r}(b)\, \xi^{b(i-i')},$$

where $W_{n,r}(b) = \frac{1}{N}\sum_{k\equiv b\ (3^r)} |\hat\lambda_k|^2 > 0$, the terms $3 \mid k$ being 0 by Step 1 and the others positive by Step 2. *Normalisation.* The left side $G_{ii'} = \sum_{j\in\mathbb{Z}/N} \lambda_{j+hi}\lambda_{j+hi'}$ is the Gram matrix of the log vectors themselves (the Gram route of the certificate computes it directly from the λ_j), and the right side is Parseval's identity for the unnormalised transform $\hat u_k = \sum_j u_j \omega^{-jk}$, whose inverse carries the factor $1/N$; this is the only place where $1/N$ enters, and it is the $1/N$ in the definition of $W_{n,r}(b)$ (the character-product route computes the right side from the $\hat\lambda_k$). Thus $G = V \operatorname{diag}(W) V^*$ with $V = (\xi^{bi})_{i<c,\ b\in(\mathbb{Z}/3^r)^\times}$, and $|\det V|^2 = \prod_{b\neq b'} |\xi^b - \xi^{b'}| = |\operatorname{disc}\Phi_{3^r}| = |\operatorname{disc}\mathbb{Q}(\zeta_{3^r})| = 3^{3^{r-1}(2r-1)}$ (Vandermonde over the c primitive 3^r-th roots of unity; the discriminant is Lemma 7.4). Therefore $(D_r^{(n)})^2 = \det G = 3^{3^{r-1}(2r-1)} \prod_b W_{n,r}(b)$, which is the second route of the certificate `certificates/p3/D3_cert_r19.json`: the two routes are independent linear-algebraic evaluations from the same log profile (λ_j) (Gram determinant; character product), and they agree to the ball radius in all 15 rows. Lean covers Step 1 (the coset-sum vanishing), the finite Fourier algebra of Step 2 (the shift rule of the twisted sum on $\mathbb{Z}/N$, the factor $\omega^{j_0 k}(\omega^{Nk/3} - 1)$ in front of $\hat\Gamma_k$, and $\omega^{Nk/3} = 1 \iff 3 \mid k$ for a primitive N-th root: `WeberP3Rel.twisted_sum_shift_diff_ne_zero`, `pow_third_eq_one_iff`), the polynomial step of Step 3 and $\varphi(3^r) = c$; what remains M with L inputs in Step 2 is only $\hat\Gamma_k \neq 0$ (the character identification, Washington's formula, Dirichlet), and Step 4 (Parseval, Vandermonde, discriminant) is M. □

Lemma 7.6 (Certified covolumes and constants at $p = 3$ [C])**.** *The JSON certificate* `certificates/p3/D3_cert_r19.json` *(format v3, generated by* `tools/gen_p3_cert_r19.py`*: every certified quantity is stored as EXACT dyadic endpoints $\pm m\cdot 2^e$, being the outward hull of two independent 4000-bit ball enclosures — the producer's, sage/r19_trackB/p3_covol_balls_r19.sage,* *and the read-only replay's, sage/r19_trackB/p3_readonly_recomputed_r19.json — widened about its centre to twice its half-width, with both enclosures and the bare hull kept per field, the radius, the precision and the rounding mode; directed 40-digit decimals for display only) encloses, for $1 \le r \le n \le 5$, $D_r^{(n)}$ by two independent linear-algebraic routes from the same log profile (λ_j) — the Gram determinant and the character product of Theorem 7.5 (quotient enclosing 1; intersection of the two enclosures stored separately) — $L_{3,n}^{\mathrm{rel}}$, $C_{n,r}^{(3)}$, $\sqrt{C_{n,r}^{(3)}}$, the comparators $G(3,r,1)$, $G(3,r,2)$ of MO16 Thm. A, and the quotients, each with radius $< 10^{-800}$; it records that $\hat\lambda_k$ encloses 0 for $3 \mid k$ and is bounded away from 0 for $3 \nmid k$ ($n \le 5$), and that $C_{n,r}^{(3)} < G(3,r,1)$, $\sqrt{C_{n,r}^{(3)}} < G(3,r,2)$ in all fifteen rows. The label [C] rests on the READ-ONLY REPLAY `scripts/verify_p3_readonly.sage` (verifier step 04f): it does not trust the JSON but rebuilds η_n, all conjugate logarithms, both routes (Gram–Schmidt; DFT with a table of powers of $e^{-2\pi i/N}$), the floor, $C_{n,r}^{(3)}$ and $G(3,s,f)$ from the definitions in ball arithmetic at 4000 bits, requires every certified interval to CONTAIN its recomputed enclosure (120/120 of the 8 load-bearing intervals of the 15 rows; overlap alone is not accepted — the r18 gate accepted overlap and 65/120 shipped intervals failed containment under it, ERRATA_R19 E19-1), re-decides the improvement from its own balls, and rejects twelve planted certificates (malformed; one route missing; wrong discriminant exponent; Schinzel floor substituted; MO13 comparator substituted; and seven that meet but do not contain the recomputed interval: shifted narrow intervals, lower endpoints moved inward, upper endpoints moved inward, Gram route alone, DFT route alone, C alone, $\sqrt{C}$ alone; ledger docs/P3_NEGCTL_LEDGER_R19.md generated from the replay's JSON). The coverage relation is re-decided on the stored recomputation by `tools/check_p3_containment.py` (step 04g); the consistency/print-string half is `tools/check_p3_cert.py` (step 04e). In particular $C_{4,4}^{(3)} \in [1.072749779 \times 10^{33}, 1.072749780 \times 10^{33}]$ and $\sqrt{C_{4,4}^{(3)}} \in [3.275285910 \times 10^{16}, 3.275285911 \times 10^{16}]$. The r16, r17 and r18 certificates (Schinzel floor; 40-digit display endpoints; overlap-era v2) are retained for the record and are rejected by the replay as deprecated.*

Theorem 7.7 (P3: layer-fixed bound in the $\mathbb{Z}_3$-tower [L-relative; F-core]). *Let $n \ge 1$, $\ell \ne 3$ an odd prime, $f \in \{1,2\}$ the order of ℓ modulo 3, $3^s \parallel \ell^2 - 1$, $r = \min(n,s)$, $c = 2 \cdot 3^{r-1}$, and $L_{3,n}^{\mathrm{rel}}$ as in Lemma 7.2. Let $D_r^{(n)} > 0$ be the covolume of Theorem 7.5 and put*

$$C_{n,r}^{(3)} = (2/\pi)^{c/2}\, \Gamma(2 + c/2)\, D_r^{(n)} / (L_{3,n}^{\mathrm{rel}})^{c}.$$

If $\ell^f > C_{n,r}^{(3)}$ then $\ell \nmid h_{3,n}/h_{3,n-1}$.

Proof. The conclusion. "$\ell \nmid h_{3,n}/h_{3,n-1}$" is read as $v_\ell(h_{3,n}) = v_\ell(h_{3,n-1})$, equivalently $\ker(N)_\ell = 0$ for the norm N of $B_{3,n}/B_{3,n-1}$ (Lemma 4.1 with $p = 3$, $\ell \ne 3$). *The prime.* Since ℓ is odd and $\ell \ne 3$, the two places where the oddness of ℓ enters are available: Lemma 7.3(iv) (the transfer of relative norm 1 to the ℓ-th root, where $\mathrm{Nr}(\varepsilon)^\ell = 1$ with $\mathrm{Nr}(\varepsilon)$ real forces $\mathrm{Nr}(\varepsilon) = 1$, `WeberP3Rel.eq_one_of_odd_pow_eq_one`) and the hypothesis of Theorem 3.8, which is stated for an odd prime. Nothing below is claimed for $\ell = 2$.

The data. Let $p = 3$, $N = 3^n$, $q = 3^{n+1}$, $\zeta = e^{2\pi i/q}$, and let σ be the generator $\zeta \mapsto \zeta^4$ of $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}(\zeta^{3^n})) \cong \mathbb{Z}/3^n$, which restricts to a generator of $G = \mathrm{Gal}(B_{3,n}/\mathbb{Q})$. The Horie unit is $\eta_n = \sin(2(1+3^n)\pi/q)/\sin(2\pi/q)$ ([10] §1; at $p = 3$ the product defining η_n has one factor), a unit of $B_{3,n}$ of relative norm 1 (Lemma 7.3). For $\alpha = \sum_{i<c_n} a_i \zeta_{3^n}^i \in \mathbb{Z}[\zeta_{3^n}]$ ($c_n = 2 \cdot 3^{n-1}$, $\zeta_{3^n} = \zeta^3$) put $\alpha_\sigma = \sum_{i<c_n} a_i \sigma^i$ and $u_\alpha = \eta_n^{\alpha_\sigma}$; for $\alpha \in \mathbb{Z}[\zeta_{3^r}] \subseteq \mathbb{Z}[\zeta_{3^n}]$ (via $\zeta_{3^r} = \zeta_{3^n}^{3^{n-r}}$) the exponents of α_σ are the multiples $3^{n-r}j$, $j < c$. The map $\alpha \mapsto H(u_\alpha)$ is additive in the coordinates a_j, and its image on $\mathbb{Z}[\zeta_{3^r}]$ is the lattice spanned by $H(\sigma^{3^{n-r}j}\eta_n)$, $j < c$, whose covolume $D_r^{(n)}$ is positive by Theorem 7.5: this is (Rank) of Theorem 3.8, for every n and r.

Saturation (the input of [10] Lemma 1.3). Since $r \le s$, the field $F = \mathbb{Q}(\zeta_{3^r})$ contains the decomposition field of ℓ in $\mathbb{Q}(\zeta_{3^n})$ (for $f = 1$ the decomposition field is $\mathbb{Q}(\zeta_{3^{\min(n,s)}})$ itself; for

$f = 2$ it is its maximal real subfield). If $\ell \mid h_{3,n}/h_{3,n-1}$, [10] Lemma 1.3 (attributed there to [9], which relays [7]; the general form with proof is [8] Prop. 1) gives a prime $\mathcal{L}$ of F above ℓ such that $u_\alpha = \eta_n^{\alpha_\sigma}$ is an ℓ-th power in $E_{3,n}$ for every $\alpha \in \ell\mathcal{L}^{-1}$. The residue degree of $\mathcal{L}$ is f, so $\ell\mathcal{L}^{-1}$ has index ℓ^{c-f} in $\mathbb{Z}[\zeta_{3^r}]$: this is (Index) with $d = f$. Let $U = U_{3,n} = \{\varepsilon \in E_{3,n} : \mathrm{Nr}_{B_{3,n}/B_{3,n-1}}\,\varepsilon = 1\}$. (Car) for $\alpha \in \ell\mathcal{L}^{-1} \setminus \ell\mathbb{Z}[\zeta_{3^r}]$: the ℓ-th root ε has $\ell H(\varepsilon) = H(u_\alpha)$ (`WeberSH.sat_root_log`) and lies in $U_{3,n}$ by Lemma 7.3(iv). (Car) for $\alpha = \ell\beta$: u_β itself carries $H(u_\alpha) = \ell H(u_\beta)$ (`WeberSH.base_branch`), and $u_\beta \in U_{3,n}$ because $\mathrm{Nr}(\eta_n^{\beta_\sigma}) = \mathrm{Nr}(\eta_n)^{\beta_\sigma} = 1$ (Lemma 7.3(iii)).

The floor. Let $\varepsilon \in U_{3,n}$ with $H(\varepsilon) \neq 0$. By definition of $U_{3,n}$, $\mathrm{Nr}_{B_{3,n}/B_{3,n-1}}(\varepsilon) = 1$; by Lemma 7.3(v) (a unit of $E_{3,n-1}$ with relative norm 1 equals 1, and $H(1) = 0$), $\varepsilon \notin E_{3,n-1}$. Hence $\varepsilon \in E_{3,n} \setminus E_{3,n-1}$ with relative norm 1, and Lemma 7.2 gives $\mathrm{ht}(\varepsilon) \geq L^{\mathrm{rel}}_{3,n}$ directly in the Euclidean height (the height of [14] Def. 2.2 is the ℓ^2-norm of the log vector; no Cauchy–Schwarz step is needed). This is (Floor) for $U = U_{3,n}$ with $L_0 = L^{\mathrm{rel}}_{3,n}$.

Conclusion. Theorem 3.8 (`WeberSH.theoremSH_contra`; `WeberP3.theoremP3_core` is the same statement with the $p = 3$ names) with $K = B_{3,n}$, $m = c$, the lattice $\ell\mathcal{L}^{-1}$, $L_0 = L^{\mathrm{rel}}_{3,n}$ and Blichfeldt's theorem ([1] Thm. II as [14] Thm. 2.7) on $\Lambda = \ell^{-1}H(u_{\ell\mathcal{L}^{-1}})$, of rank c and covolume $D_r^{(n)}/\ell^f$ in its span, yields $\ell^f \leq (2/\pi)^{c/2}\Gamma(2 + c/2)D_r^{(n)}/(L^{\mathrm{rel}}_{3,n})^c = C^{(3)}_{n,r}$. Contrapositively, $\ell^f > C^{(3)}_{n,r}$ implies $\ell \nmid h_{3,n}/h_{3,n-1}$.

Remark (what changes relative to [10], [14]). The saturation input is that of [10]; the floor is that of [14] Lemma 2.5(2), applicable because $\mathrm{Nr}\,\eta_n = 1$ transfers to the saturation root (Lemma 7.3); the change is the convex body: [10] §4–5 and [14] §4 apply Minkowski's or Blichfeldt's theorem to an ideal lattice in coordinates and bound the height of δ_n^α through the Mahler measure, which produces the factors $c!$ resp. $((c+2)/2)!\,(\sqrt{2\pi}/(3^{3/4}\log(\cdots)))^c$ in $G_1(3, s, f)^f$ and $G(3, s, f)^f$; here Blichfeldt's theorem acts on the log lattice with its exact covolume $D_r^{(n)}$ (Theorem 7.5), and the floor enters through $(L^{\mathrm{rel}}_{3,n})^c$ with $L^{\mathrm{rel}}_{3,n}$ growing like $\sqrt{3^n}$. $\square$

Corollary 7.8 ($n = 4$ [C + M]). *If $\ell \equiv 1 \pmod{81}$ and $\ell > 1.0728 \times 10^{33}$, or $\ell \equiv -1 \pmod{81}$ and $\ell > 3.2753 \times 10^{16}$, then $\ell \nmid h_{3,4}/h_{3,3}$. In the classes $81 \,\|\, \ell \mp 1$ these layer-fixed thresholds are below the uniform-in-n bound of Morisawa and Okazaki, [14] Thm. A (3.7111×10^{37}, resp. 6.0919×10^{18}), by the factors 3.4594×10^4, resp. 1.8599×10^2.*

Proof. At $n = 4$ we have $N = 81$, $s \geq 4$ means $81 \mid \ell^2 - 1$, i.e. $\ell \equiv \pm 1 \pmod{81}$, so $r = 4$, $c = 54$, and $f = 1$ for $\ell \equiv 1$, $f = 2$ for $\ell \equiv -1 \pmod{81}$; $L^{\mathrm{rel}}_{3,4} = 9\log\frac{3^{40/81} + \sqrt{3^{80/81} + 4}}{2} = 7.0107178918636975\ldots$. The certificate `certificates/p3/D3_cert_r19.json` is the outward hull, widened to twice its half-width, of two independent 4000-bit ball enclosures of every quantity below: the producer's (the 81 logarithms $\log|\sigma^t\eta_4|$ are balls, the 54×54 Gram matrix and its determinant are ball operations, and independently the discrete Fourier transform of the log vector and the character product of Theorem 7.5 Step 4; the two enclosures of $D_4^{(4)}$ have radius below 10^{-1000} and their quotient encloses 1) and the read-only replay's `scripts/verify_p3_readonly.sage`, which rebuilds the same quantities from the definitions by Gram–Schmidt orthogonalisation and a table of powers of $e^{-2\pi i/N}$; the certificate is generated by `tools/gen_p3_cert_r19.py`, and the replay requires every certified interval to contain its own recomputation (coverage checker `tools/check_p3_containment.py`). It encloses

$$C^{(3)}_{4,4} = \frac{(2/\pi)^{27}\,\Gamma(29)\,D_4^{(4)}}{(L^{\mathrm{rel}}_{3,4})^{54}} \in [1.072749779 \times 10^{33}, 1.072749780 \times 10^{33}],$$

$$\sqrt{C^{(3)}_{4,4}} \in [3.275285910 \times 10^{16}, 3.275285911 \times 10^{16}].$$

Hence $\ell \equiv 1 \pmod{81}$ and $\ell > 1.0728 \times 10^{33}$ give $\ell^f = \ell > C^{(3)}_{4,4}$, and $\ell \equiv -1 \pmod{81}$ and $\ell > 3.2753 \times 10^{16}$ give $\ell^f = \ell^2 > C^{(3)}_{4,4}$; Theorem 7.7 applies. The same certificate gives the rows $n \le 5$ of the comparison table in the $\mathbb{Z}_3$ section of the paper. For the comparison: the uniform-in-n bound of Morisawa and Okazaki, [14] Thm. A, which for $p = 3$, $s = 4$ reads $\ell > G(3,4,f)$ with $G(3,s,f) = \big((\sqrt{2\pi}/(3^{3/4}\log\frac{3^{40/81}+\sqrt{3^{80/81}+4}}{2}))^c\,((c+2)/2)!\big)^{1/f}$, $c = 2\cdot 3^{s-1}$, i.e. $G(3,4,1) \ge 3.7111\times 10^{37}$ and $G(3,4,2) \ge 6.0919\times 10^{18}$ (the same certificate encloses G; its value $G(3,3,2) = 42407.49\ldots$ agrees with the table of [14] Example 1.6, "4.3×10^4"). In the class $\ell \equiv 1 \pmod{81}$ with $81 \parallel \ell - 1$ (the set $D(3,4,1)$ of [14]) the present threshold is smaller by the factor $G(3,4,1)/C^{(3)}_{4,4} \ge 3.4594\times 10^4$, and in $D(3,4,2)$ ($\ell \equiv -1 \pmod{81}$, $81 \parallel \ell+1$) by $G(3,4,2)/\sqrt{C^{(3)}_{4,4}} \ge 1.8599\times 10^2$. The comparison convention is that of the paper's comparison section (layer-fixed versus uniform in n): [14] is uniform in n (and its constant for $s \ge 5$ is $G(3,s,f)$, larger), the present bound is for the layer $n = 4$ only; [10] Thm. 0.3 ($2^{c/2}c! = 3.098\times 10^{79}$ at $s = 4$) and Thm. A ($G_1(3,4,f)^f = 7.63\times 10^{101}$) are listed in that table for the record, not as the comparator. $\square$

Chapter 8

Certificates

Definition 8.1 (Verifier specification [C])**.** scripts/family_verify.sage (r11, sha256 0f38bb0d...; unchanged in r12–r13): on input a witness file it recomputes the factorisation of $x^m + 1$ over $\mathbb{F}_\ell$, asserts the header fields, and accepts a line (f, route, a) only if f is a recomputed factor, $f \cdot a \equiv 0 \bmod (\ell, x^m{+}1)$, $a \notin \ell R_n$, some log coordinate is certified nonzero, and (RHO) $\mathrm{ht}(\ell^{-1} H_n a) < L_n$ resp. (T) $\sum_j e^{2y_j} < \bar{T}_n$ hold with the ball's upper endpoint, where $\bar{T}_n = 17 \cdot 2^n$ for $n = 2$ and $33 \cdot 2^n$ for $n \geq 3$ is recomputed from n (Lemma 2.4; a header claiming another threshold is rejected), the claimed value agreeing to 10^{-6}. Verdict EXCLUDED iff every component has an accepted line and no line fails.

Theorem 8.2 (Certificate core: direct KY chain [F])**.** *If $\ell \mid k_n$ implies a saturated component, if a RHO witness (nonzero vector of height $< L$) and a T witness (nonzero vector with $T < \bar{T}$) each refute saturation, and if every component carries one of them, then $\ell \nmid k_n$; `exclusion_of_checker_spec` states the verifier specification "EXCLUDED $\Rightarrow$ every component carries a witness" as an explicit hypothesis (the implementation is NOT verified).*

Proof. Assume $\ell \mid k_n$. By the first hypothesis some component f is saturated. By the covering hypothesis f carries a RHO witness or a T witness. In the first case the RHO hypothesis says a RHO witness on f refutes saturation of f; in the second case the T hypothesis does the same. Either way f is not saturated, a contradiction; so $\ell \nmid k_n$ (`exclusion_chain_direct`, a proof by cases with no axiom). The two refutation hypotheses are themselves discharged from the floors: if saturation of f forces $L \leq \mathrm{ht}(v)$ (resp. $\bar{T} \leq T(v)$) for every nonzero v on f, then a nonzero v with $\mathrm{ht}(v) < L$ (resp. $T(v) < \bar{T}$) is incompatible with saturation (`rho_witness_refutes_saturation`, `t_witness_refutes_saturation`: $a \leq b$ and $b < a$ cannot both hold). Finally `exclusion_of_checker_spec` assembles these with the specification hypothesis "the checker's EXCLUDED verdict implies that every component has a nonzero vector with ht $< L$ or one with $T < \bar{T}$": feeding that hypothesis and the two floors into the chain gives $\ell \nmid k_n$ from the verdict. The correspondence between the verdict of the actual script and this predicate is the specification of Definition 8.1; it is a hypothesis here, not a Lean fact. The Lean declaration(s) named above are this proof verbatim; `#print axioms` within std-3 (lean/README_lean.md). $\square$

Lemma 8.3 (The piecewise trace threshold $\bar{T}_n$ [F])**.** *$\bar{T}_n := 17 \cdot 2^n$ for $n = 2$ and $33 \cdot 2^n$ for $n \geq 3$ (lean/WeberCertFloor.lean, r13, E13-1). Then $\bar{T}_2 = 68$, $\bar{T}_7 = 4224$, $\bar{T}_n \leq 33 \cdot 2^n$ for all n with $\bar{T}_2 < 33 \cdot 2^2$, and the T-refutation of Theorem 8.2 holds at* bar $= \bar{T}_n$.

Proof. $\bar{T}_n$ is defined by cases on n: $\bar{T}_2 = 17 \cdot 2^2 = 68$ and $\bar{T}_n = 33 \cdot 2^n$ for $n \geq 3$, the two literature trace floors on $RE_n^+ \setminus \{\pm 1\}$ ([15] Prop. 6.6 for every $n \geq 2$; [6] Thm. 2.3 for $n \geq 3$). The values $\bar{T}_2 = 68$ and $\bar{T}_7 = 4224$ are read off the definition. For every n, $\bar{T}_n \leq 33 \cdot 2^n$ since $17 < 33$, with strict inequality exactly at $n = 2$; so a T line accepted under the piecewise threshold is accepted under the uniform one, and the converse fails at $n = 2$ for $68 \leq T < 132$, which is why the uniform constant cannot be used in Theorem 8.4. The refutation of saturation by a T witness is then the abstract statement of Theorem 8.2 instantiated at bar $= \bar{T}_n$: if saturation of f forces $T(v) \geq \bar{T}_n$ on every nonzero v (the floor hypothesis), a nonzero v with $T(v) < \bar{T}_n$ contradicts it. □

Theorem 8.4 (Certificate soundness [C+L, F-core]). *If the read-only verifier of Chapter 8returns* EXCLUDED *for (n, ℓ), then $\ell \nmid k_n$.*

Proof. Suppose $\ell \mid k_n$. [6] Prop. 4.1 gives a saturated component f_0 (Lemma E). The accepted line on f_0 names a vector $a \in L_{f_0} \setminus \ell R_n$ with $v = \ell^{-1} H_n a \neq 0$; by Lemma E, v is the log vector of $r_a \in RE_n^+$, $r_a \neq \pm 1$. A RHO line asserts $|v| < L_n$, contradicting Lemma A with [14] Lemma 2.5(1); a T line asserts $\sum_j \sigma^j(r_a)^2 = \sum_j e^{2y_j} < \bar{T}_n$, where $\bar{T}_n = 33 \cdot 2^n$ for $n \geq 3$ and $\bar{T}_n = 17 \cdot 2^n$ for $n = 2$ (the verifier recomputes $\bar{T}_n$ from n), contradicting the trace floor $\mathrm{Tr}(r_a^2) \geq \bar{T}_n$ of [6] Thm. 2.3 for $n \geq 3$ and of [15] Prop. 6.6 for $n = 2$ (the uniform constant $33 \cdot 2^n$ would not give a contradiction at $n = 2$ for $68 \leq T < 132$). Hence $\ell \nmid k_n$. □

Lemma 8.5 (KY1000 certificates [C]). *For every $\ell \in \mathcal{P}$ the witness file for ℓ (one file per prime, `certificates/family/KY1000/`, SHA-256 in the manifest; the paper's Appendix F (artifact index; the paths are in `MANIFEST_SHA256.txt`)) is accepted by the frozen family verifier (SHA-256 `0f38bb0d`...) with verdict EXCLUDED: the verification ledger reads EXCLUDED* 1000/1000*; nine planted corruptions of a passing file are all REJECTED and the untouched $n = 2$ file is accepted (negative-control log and ledger, the paper's Appendix F (artifact index; the paths are in `MANIFEST_SHA256.txt`)).*

Note. Certificate id FAMILY_KY1000: `certificates/family/KY1000/witness_n7_l*.txt` (1000 files, SHA-256 in `MANIFEST_SHA256.txt`); clean run `sage/family_ky1000_r11_clean/VERIFY_LEDGER.txt`: EXCLUDED 1000/1000; negative controls 9/9 rejected and the $n = 2$ positive control accepted (`sage/negctl_r13/negctl_r13.log`; `certificates/negctl/negctl_ledger_r14.json`).

Lemma 8.6 (KY1000 target list [C]). *Certificate id KY1000_TARGET: `sage/family_ky1000/KY1000_primes.txt` has 1000 distinct ascending primes, all $\equiv 65$ (128), all $> 10^9$, elementwise equal to the first 1000 qualifying primes regenerated from 10^9, equal to the set*

$$\{\ell \text{ prime} : 10^9 < \ell \leq 1001287361,\ \ell \equiv 65\ (128)\},$$

and equal to the witness-file set; checker `scripts/verify_ky1000_target.py` (verifier step 03b; 5 planted corruptions rejected, step 03c).

Theorem 8.7 (Family [C+L, F-core]). *Let $\mathcal{P}$ be the set of the first* 1000 *primes ℓ with $\ell > 10^9$ and $\ell \equiv 65 \pmod{128}$; $\min \mathcal{P} = 1000000321$, $\max \mathcal{P} = 1001287361$, and $\mathcal{P}$ is the list of [6] §4.2; equivalently, $\mathcal{P} = \{\ell \text{ prime} :\ 10^9 < \ell \leq 1001287361,\ \ell \equiv 65 \pmod{128}\}$. For every $\ell \in \mathcal{P}$ one has $\ell \nmid h_7/h_6$.*

Proof. The finite set $\mathcal{P}$. Define $\mathcal{P}$ as the set of the first 1000 primes $\ell > 10^9$ with $\ell \equiv 65 \pmod{128}$, i.e. the 1000 smallest such primes. This is a finite, explicitly enumerable set: the primes $\equiv 65 \pmod{128}$ above 10^9 form an increasing sequence, and $\mathcal{P}$ is its initial segment of length 1000. Its largest element is 1001287361, so $\mathcal{P} = \{\ell \text{ prime} : 10^9 < \ell \leq 1001287361,\ \ell \equiv 65 \pmod{128}\}$; the two descriptions agree because the second set has exactly 1000 elements and contains the first. The endpoints coincide with those printed in [6] §4.2.

Target-list certificate (claim KY1000_TARGET, label C). The shipped list (`sage/family_ky1000/KY1000_primes.txt`) is checked, by a script with no dependence on the generator, to have 1000 entries, distinct and ascending, every entry prime (deterministic primality test for this size), every entry $\equiv 65 \pmod{128}$ and $> 10^9$; the script then regenerates the first 1000 qualifying primes from 10^9 by a sieve and compares elementwise, and finally compares the list with the set of primes in the witness-file names. The target-list checker also rejects five planted corruptions of the list (one entry altered to a composite in the wrong class; one qualifying prime omitted and the list padded with the 1001st; the last entry duplicated; two neighbours swapped; one prime replaced by the next qualifying prime beyond 1001287361, which only the interval form catches; verifier steps 03b/03c, a control set separate from the nine witness-file controls). This discharges the identification of the shipped list with the set $\mathcal{P}$ defined above.

Witness certificates (Lemma 8.5, label C). For every $\ell \in \mathcal{P}$ there is one witness file (SHA-256 in the manifest). The read-only verifier `scripts/family_verify.sage` (specification in the paper's certificate section; frozen; SHA-256 `0f38bb0d`...) accepts each of them with verdict EXCLUDED: the verification ledger records EXCLUDED 1000/1000, every one of the $32 = 2^{7-2}$ components of every prime carrying an accepted T certificate (32000 T lines) and 31987 of them also an accepted RHO line; the largest accepted T value is $4164.4897 < \bar{T}_7 = 4224$. Nine planted corruptions of a passing file (wrong factor, wrong vector, vector in ℓR_n, altered threshold, altered header fields, ...) are REJECTED and the untouched $n = 2$ file is accepted (`sage/negctl_r13/negctl_r13.log`; ledger `certificates/negctl/negctl_ledger_r14.json`).

From EXCLUDED to $\ell \nmid h_7/h_6$ (Theorem 8.4, C+L, F-core). By Theorem 8.4, EXCLUDED for $(7, \ell)$ implies $\ell \nmid k_7 = h_7/h_6$: each accepted line is a nonzero vector of the component lattice refuting saturation of that component (RHO: height below the floor L_7; T: trace of the square below $\bar{T}_7$, [6] Thm. 2.3), and saturation of some component is forced by $\ell \mid k_7$ ([6] Prop. 4.1). Both refutations are instances of the same discrete core as Theorem A (`WeberCert.exclusion_chain_direct`); the literature inputs are those of the trust table of the paper (KY Eq. (17), KY Prop. 4.1, KY Def. 2.1, MO16 Lemma 2.5(1), Blichfeldt) plus the trace floor KY Thm. 2.3 / MO3 Prop. 6.6.

Scope. Nothing is claimed for primes outside $\mathcal{P}$, for other residue classes, or for $n \neq 7$; [6] had excluded these 1000 primes conditionally on their Conjecture 2.2, which is not used here. □

Chapter 9

Experiments (used in no theorem)

Lemma 9.1 (Class 1 centered/local scan [E] (intentional non-theorem node, used by nothing))**.** *At* $\ell = 1000000513$*, the centered representatives with local* $\pm\ell$*-coordinate descent give no T or RHO witness on any of the 64 components. Not a full CVP; no theorem uses this. Log: sage/r9_tscan64_1000000513.log.*

Lemma 9.2 (Classes 63/127 near-misses [E] (intentional non-theorem node, used by nothing))**.** *Exact SVP at* $\ell = 10000003199$*:* $\operatorname{ht}/L_7 \in [1.009, 1.011]$ *on the component minima. Log: sage/r10_escalate_l10000003199.log; certificates/experiments/family_extra/.*

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